

# 115.109 Introduction to Business Analytics

## Topic 3: Industry Trends Averages



**Dr Andrew Murphy**  
(Please contact me via Communication Tools in Stream)

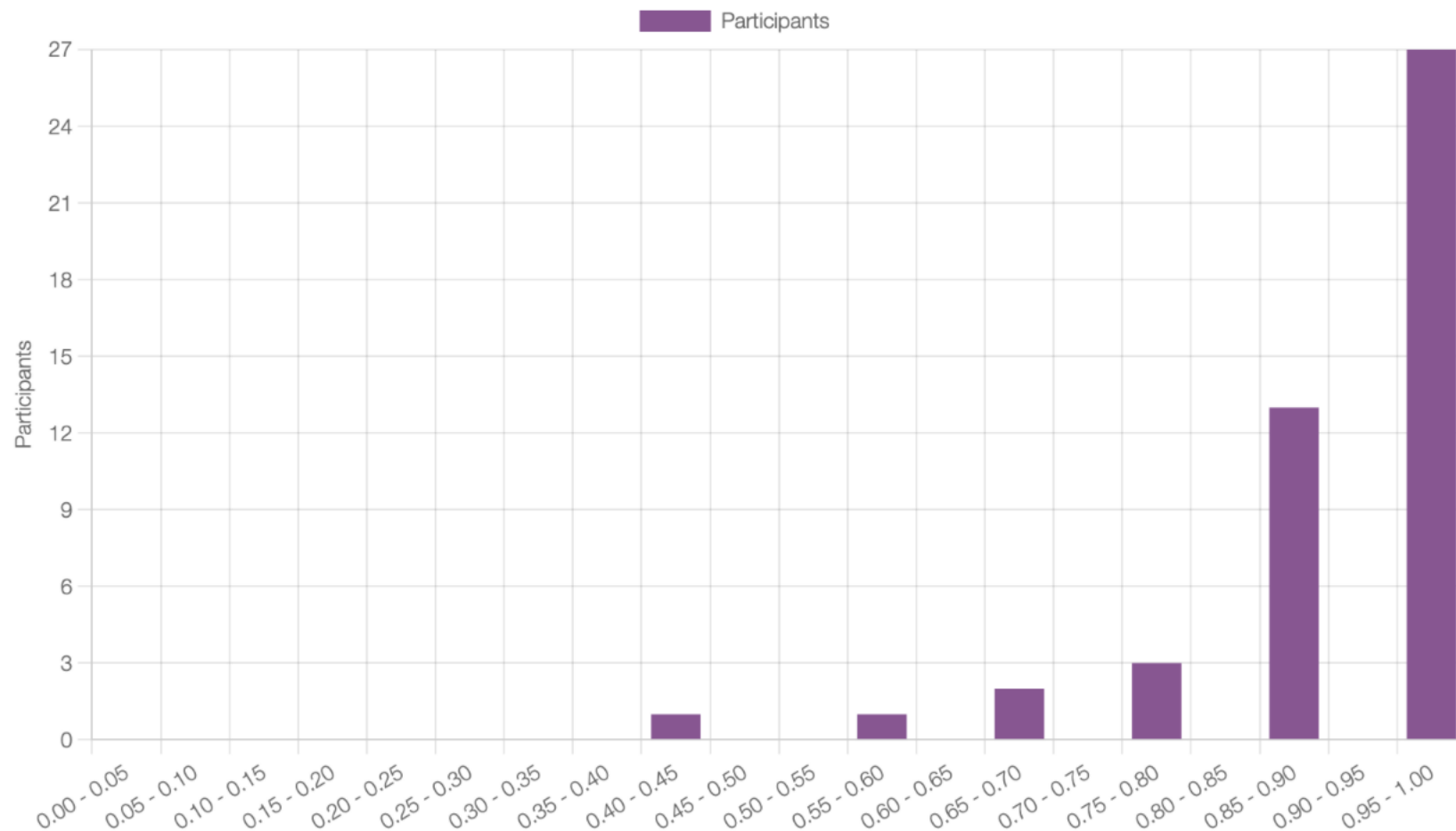
| Week (starting) | Business Problem                       | Data & Analysis                                             | Chapter (Black et al) | Assessments     |
|-----------------|----------------------------------------|-------------------------------------------------------------|-----------------------|-----------------|
| 1 (13/7)        | Understanding Data: discounts & margin | Calculating data in Excel; GST, discounts and Margins       | 1                     | Lab1            |
| 2 (20/7)        | Operational Monitoring: Dashboards     | Tabulating & Graphing                                       | 2                     | Lab2            |
| 3 (27/7)        | Industry trends                        | Statistics NZ: retail survey<br>Descriptive stats: averages | 3                     | Lab3            |
| 4 (3/8)         | Customer trends                        | Statistics NZ: census<br>Descriptive stats: variation       | 3                     | Lab4            |
| 5 (10/8)        | Transaction data: product sales        | Transaction data sources.<br>Sampling                       | 7                     | Lab5            |
| 6 (17/8)        | Loyalty cards                          | Loyalty card schemes.<br>Probability; proportions, $\chi^2$ | 4, 12                 | Lab6<br>Test 1  |
| Mid             | Semester                               | Study                                                       | Period                |                 |
| 7 (7/9)         | Segmentation                           | Statistics NZ: census. Normal distribution; Means           | 6, 8                  | Lab7            |
| 8 (14/9)        | Determining difference (1)             | Means; ANOVA                                                | 9, 10, 11             | Lab8            |
| 9 (21/9)        | Associating and Predicting             | Explaining relationships: Correlation & regression          | 13                    | Lab9            |
| 10 (28/9)       | Determining Difference (2)             | Explaining relationships (continued)                        | 14                    | -               |
| 11 (5/10)       | Trends over time                       | Time series, Indices<br>Unstructured data: social media     | 15                    | Lab11<br>Test 2 |
| 12 (12/10)      | Corporate reporting                    | Finance Dashboards reporting                                |                       | Report          |

# Assessments Schedule

| 2026                                                                                   | Assessment                       | Lab 1&2  | Lab 3&4  | Lab 5&6  | Test 1               | Lab 7&8  | Lab 9,11 | Test 2               | Report   |
|----------------------------------------------------------------------------------------|----------------------------------|----------|----------|----------|----------------------|----------|----------|----------------------|----------|
| Week                                                                                   | 13-Jul                           | O: 13.07 |          |          |                      |          |          |                      |          |
| commencing                                                                             | 20-Jul                           |          | O: 20.07 |          |                      |          |          |                      |          |
| Monday:                                                                                | 27-Jul                           |          |          |          |                      |          |          |                      |          |
|                                                                                        | 03-Aug                           | D: 03.08 |          |          |                      |          |          |                      |          |
|                                                                                        | 10-Aug                           |          |          | O: 10.08 |                      |          |          |                      |          |
|                                                                                        | 17-Aug                           |          | D: 17.08 |          | O: 14.08<br>C: 21.08 |          |          |                      | O 17.08  |
|                                                                                        | <i>Mid-Semester Study Period</i> |          |          | D:31.08  |                      |          |          |                      |          |
|                                                                                        | 07-Sep                           |          |          |          |                      | O: 07.09 |          |                      |          |
|                                                                                        | 14-Sep                           |          |          |          |                      |          |          |                      |          |
|                                                                                        | 21-Sep                           |          |          |          |                      |          | O: 21.09 |                      |          |
|                                                                                        | 28-Sep                           |          |          |          |                      |          |          |                      |          |
|                                                                                        | 05-Oct                           |          |          |          |                      | D: 05.10 |          | O: 02.10<br>C: 09.10 |          |
|                                                                                        | 12-Oct                           |          |          |          |                      |          | D: 19.10 |                      | D: 19.10 |
|                                                                                        |                                  |          |          |          |                      |          |          |                      |          |
| O=opens (available from); D=due (without late penalty); C=closes (not available after) |                                  |          |          |          |                      |          |          |                      |          |

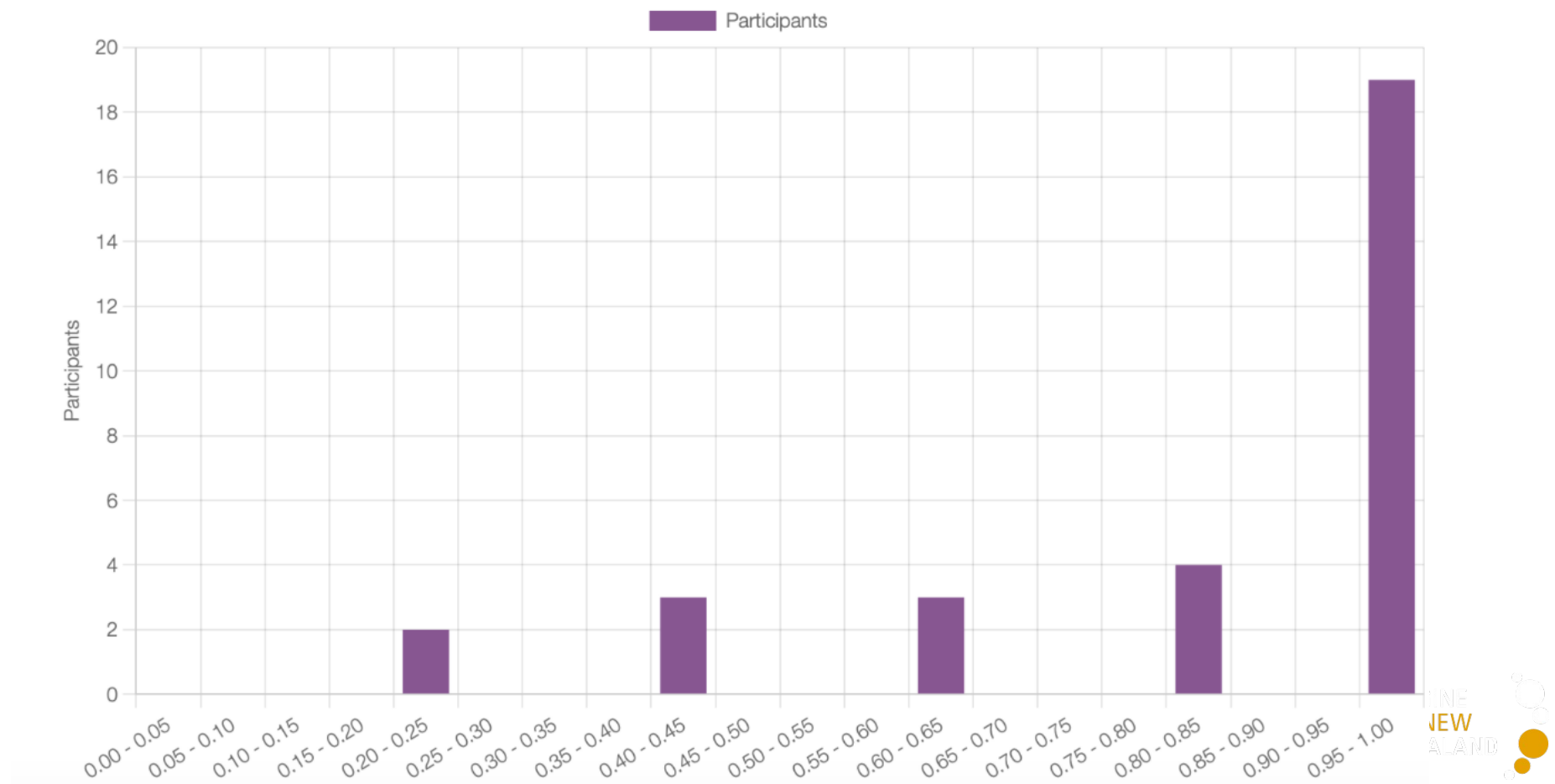
# Labs 1&2: as of 27/7

- 27 submitted lab file
- 46 completed Lab 1 Red Box Quiz ( $\bar{x}$ : 0.92)



# Labs 1&2: as of 27/7

- 27 submitted lab file
- 46 completed Lab 1 Red Box Quiz ( $\bar{x}$ : 0.92)
- 31 completed Lab 2 Red Box Quiz ( $\bar{x}$ : 0.83)



# Group Information Survey

## Contents

- ↓ [Assessment Overview](#)
- ↓ [Assessment Planner](#)
- ↓ [Formal Requirements to pass this course](#)
- ↓ [Assessment resources](#)
- ↓ [Assessment 1 information: Computer Exercises](#)
- ↓ [Missing AI Declarations?](#)
- ↓ [Labs 1 & 2](#)
- ↓ [Labs 3 & 4](#)
- ↓ [Assessment 2 & 3 information: Online Tests](#)
- ↓ [Optional Reviews](#)
- ↓ [Assessment 4 information](#)
- ↓ [Analytical Report](#)
- ↓ [Report Groups](#)



### QUESTIONNAIRE

Established group identifier: closes 10 August

1 group so far

Enter the Student IDs of ALL of the group members (either 2 or 3 members per group) IF you have already formed a group, or know of other students in the class you would like to form a group with.

If you have not yet formed a group, I will allocate you - complete the [Group Information Survey](#) to assist this.

Available now



### QUESTIONNAIRE

Group Information Survey: closes 10 August

**If you have not already formed a group, please complete this survey by Monday August 10**

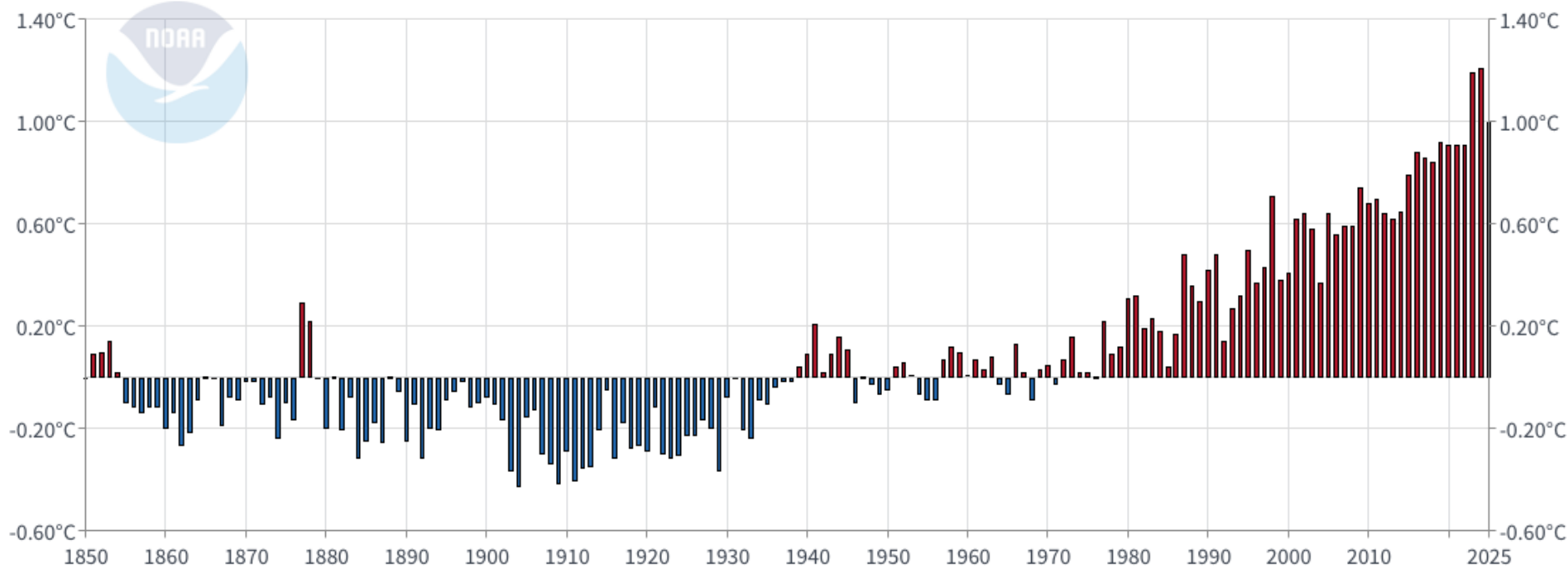
This survey will help me generate teams of similar learners for Assignment 4 (group report). All information in the survey is confidential and is used solely for the purpose of organising teams for this assessment.

Teams created using the survey information tend to be better organised than teams randomly assigned.

Items with black \* in the question will be shared publicly with your teammates (name, email and city only). The red asterisk next to the question number indicates a required question. This information is not shared.

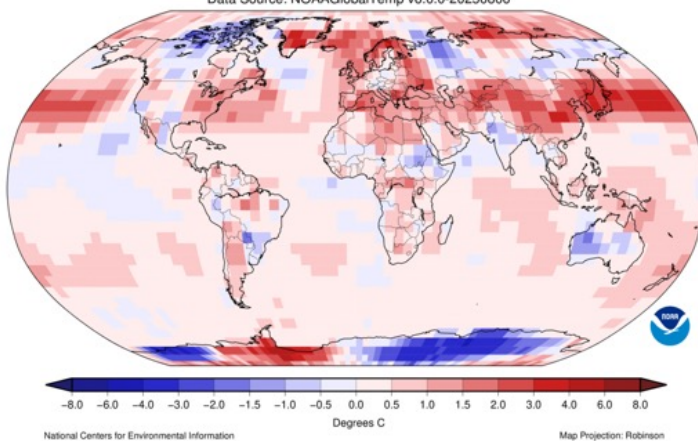
# Graph Example: Bar chart

<https://www.ncei.noaa.gov/access/monitoring/monthly-report/global/202507>



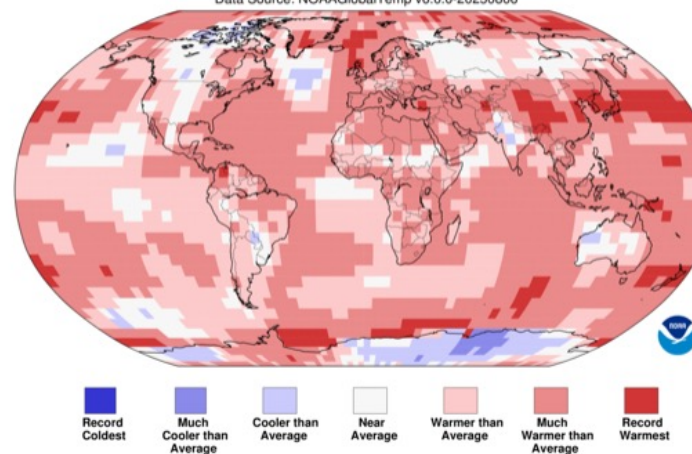
Land & Ocean Temperature Departure from Average Jul 2025  
(with respect to a 1991-2020 base period)

Data Source: NOAA GlobalTemp v6.0.0-20250806



Land & Ocean Temperature Percentiles Jul 2025  
NOAA's National Centers for Environmental Information

Data Source: NOAA GlobalTemp v6.0.0-20250806



Arctic Antarctic

Start month: January

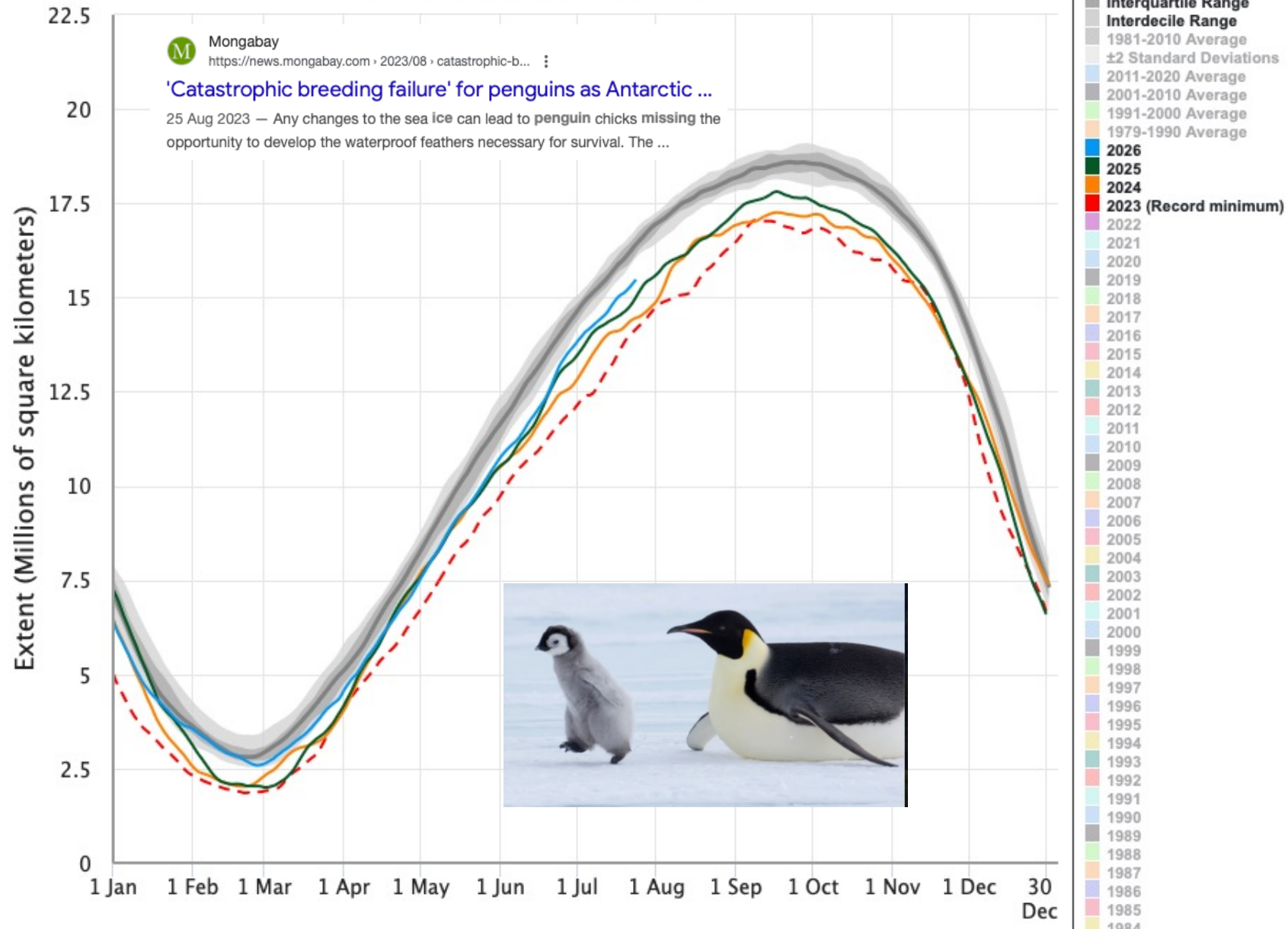
Colors: Classic

Invert colors ☐

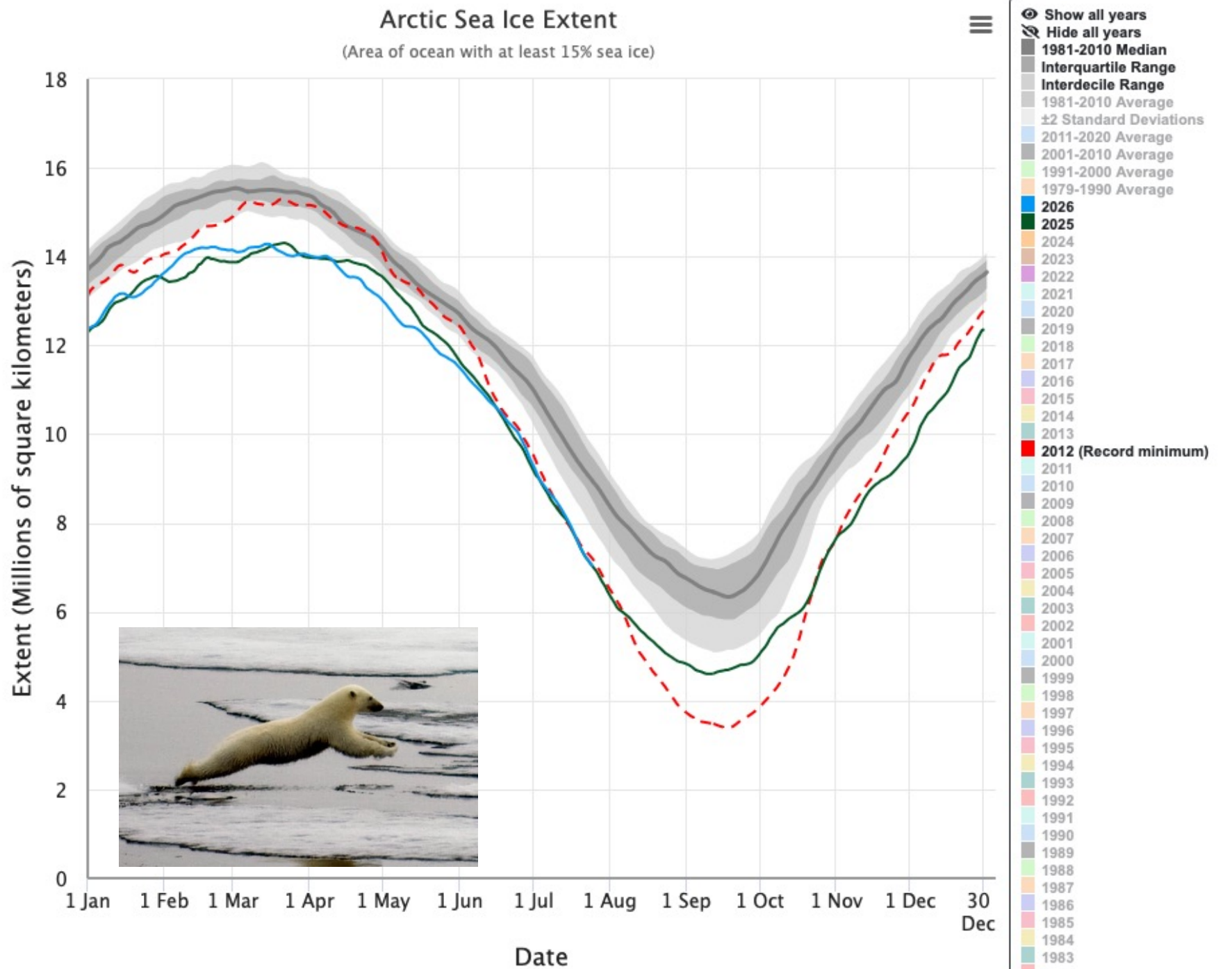
Vertical gridlines ☒

## Antarctic Sea Ice Extent

(Area of ocean with at least 15% sea ice)



<https://nsidc.org/sea-ice-today/sea-ice-tools/charctic-interactive-sea-ice-graph>



# House Prices: NZ

<https://www.interest.co.nz/charts/real-estate/median-price-reinz>

Select chart tabs

NZ total

Median price



Be careful when you see other house price reports - they often report 'average', and this can be moved more easily by a few outlier prices especially when volumes are low...

'median price' is of all types and specification of houses sold. The range will include small shoebox apartments all the way up to palatial mansions. The REINZ series makes no distinction. In fact, nationally, the 'middle price' might be a house in the Waikato one month, and one in Wellington the next, for example. There is no filter that means it will always be (for example) a 3 bedroom, 1000m2 section, or in fact any standard.

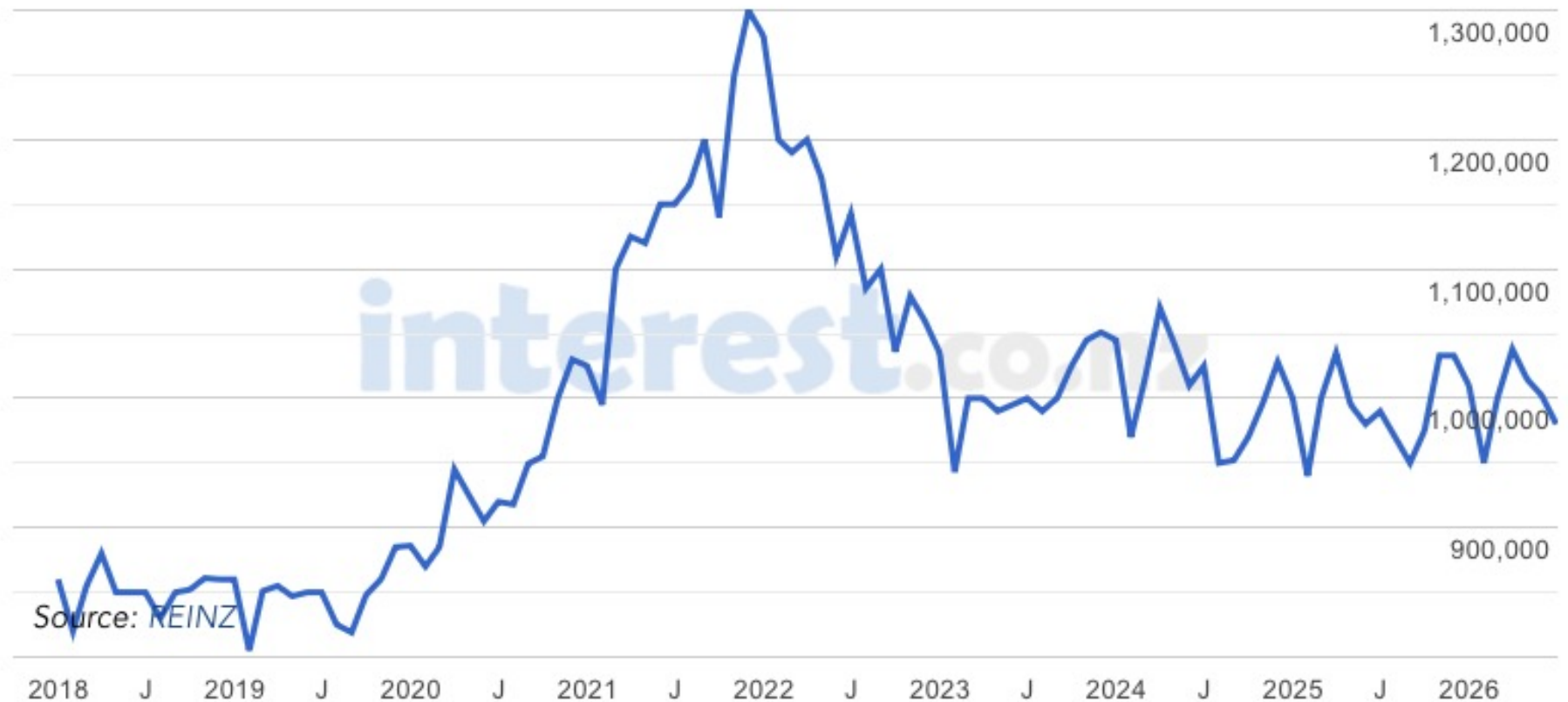
# House Prices: Auckland

<https://www.interest.co.nz/charts/real-estate/median-price-reinz>

Select chart tabs

Auckland

Median price



Source: REINZ

# House Price Growth: Auckland

<https://www.interest.co.nz/charts/real-estate/median-house-price-growth>

## Select chart tabs

Auckland %

## Median price



Source: REINZ

# Barfoots data 2002-2026

<https://www.interest.co.nz/charts/real-estate/barfoot-auckland>

## Select chart tabs

average price growth % year on year

✓ average price growth % year on year

average price

number sold

new listings

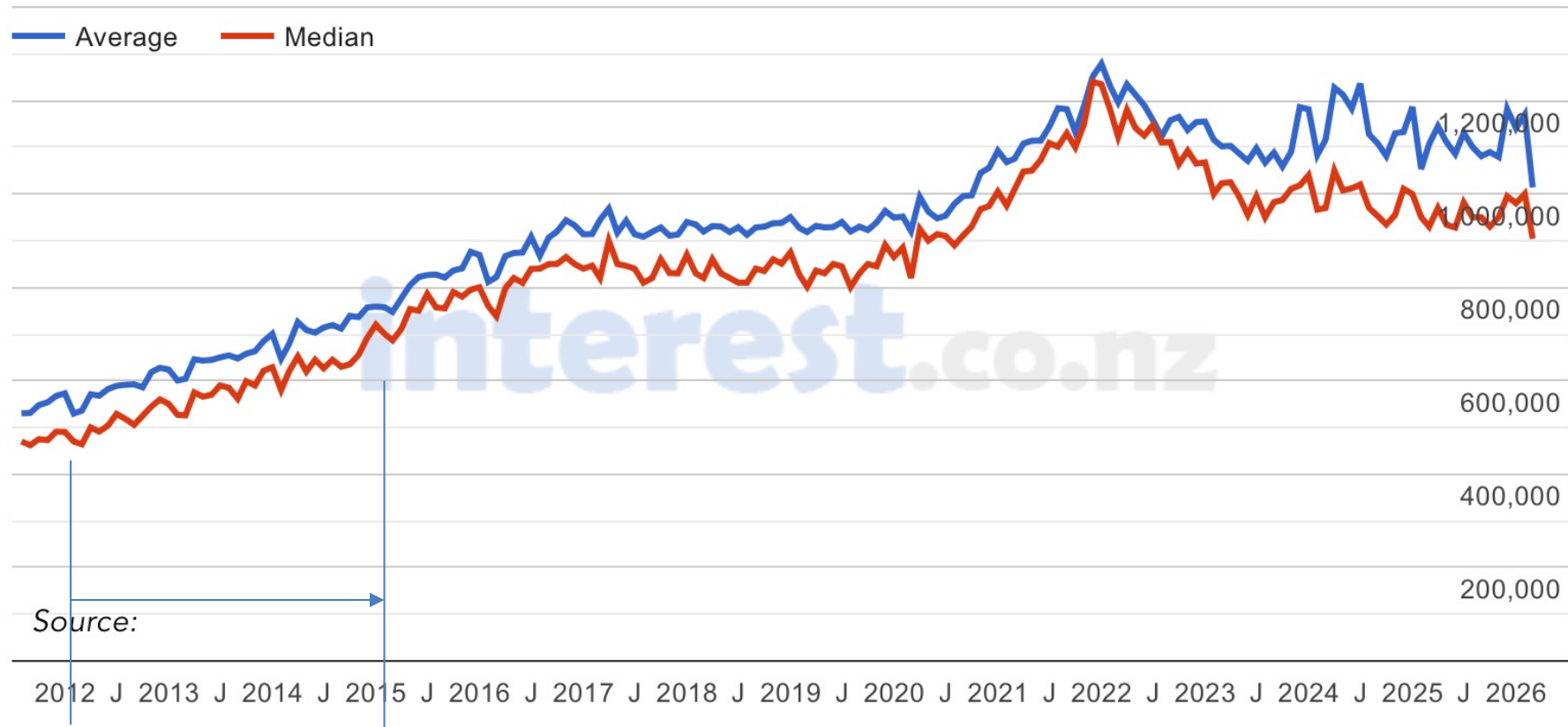


Average here is the mean price

# House Prices: Auckland

<https://www.interest.co.nz/charts/real-estate/barfoot-auckland>

## average price



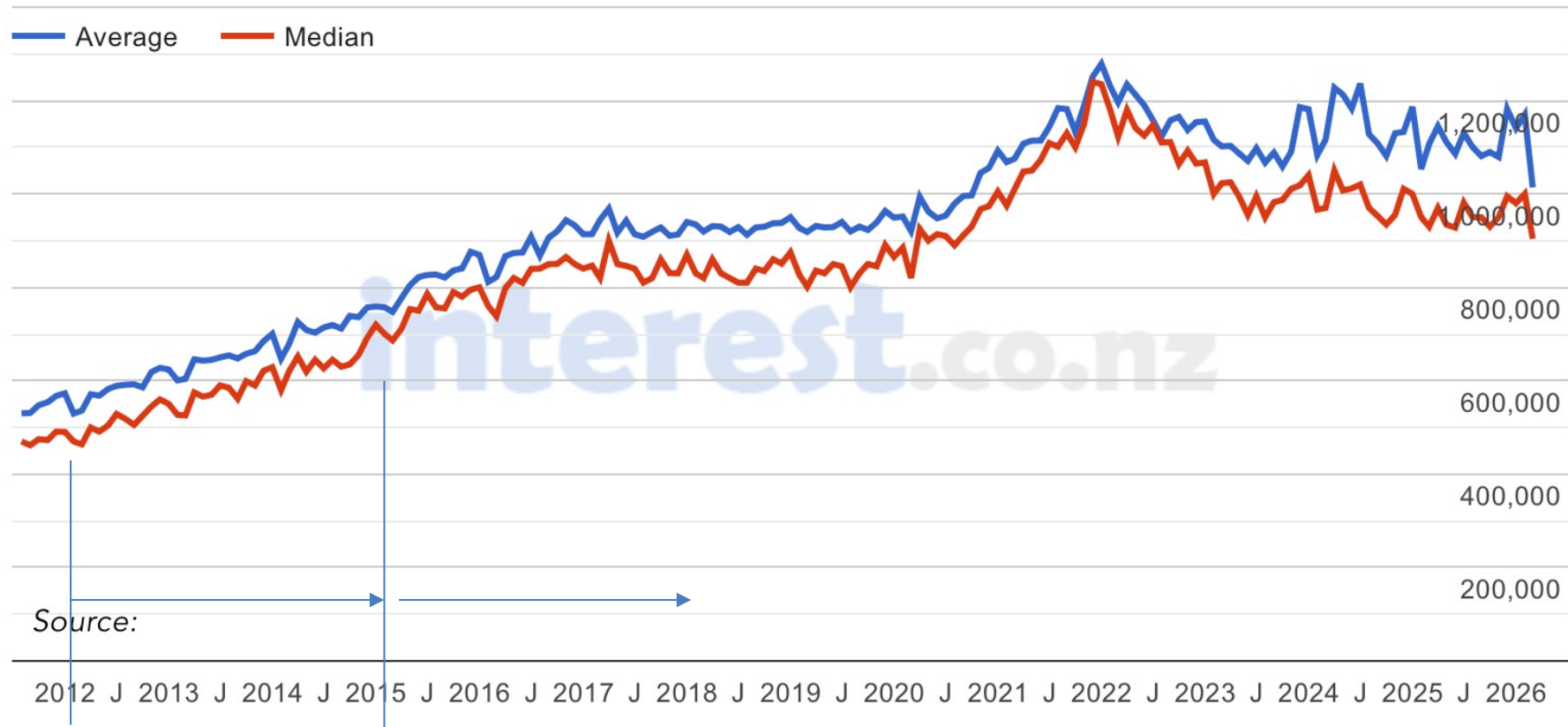
Q: Median price was \$500,000 in March 2012 and increases by 25% over 3 years: what was it in March 2015?

$$500k \times 1.25 = \$625k$$

# House Prices: Auckland

<https://www.interest.co.nz/charts/real-estate/barfoot-auckland>

## average price



Q: Median price was \$500,000 in March 2012 and increases by 25% over 3 years: what was it in March 2015?

$$500k \times 1.25 = \$625k$$

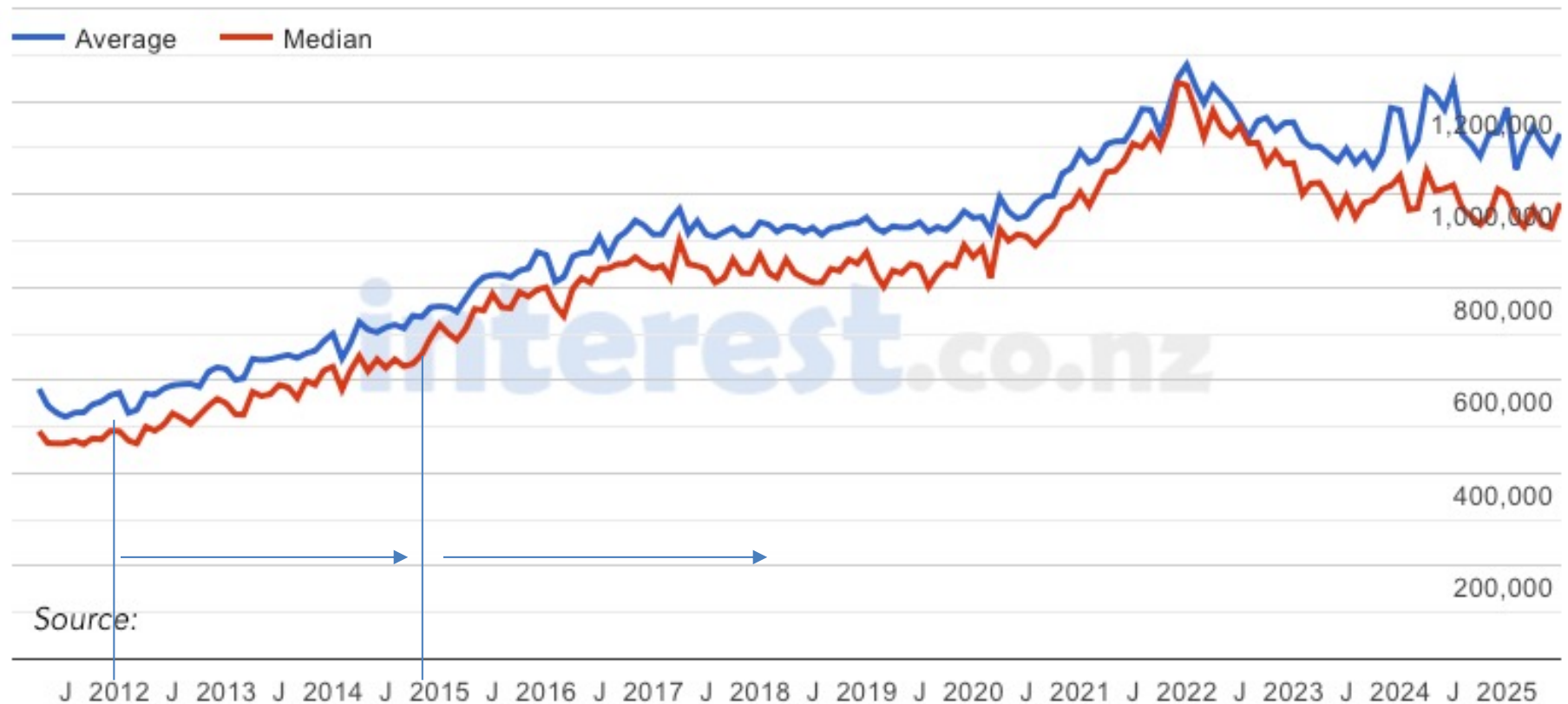
If the median price then fell by 25% over 3 years, what would it have been by 2018?

$$625k \times 0.75 = \$469k$$

# House Prices: Auckland

<https://www.interest.co.nz/charts/real-estate/barfoot-auckland>

## average price



Q: Median price was \$500,000 in March 2012 and increases by 25% over 3 years: what was it in March 2015?

$$500k \times 1.25 = \$625k$$

If the median price then fell by 25% over 3 years, what would it have been by 2018?

$$625k \times 0.75 = \$469k$$

# Areas of Statistics

1. **Descriptive Statistics:** summarising data (mean, max, min) without making an interpretation (or inferences)
2. **Exploratory Data Analysis:** use of statistical and graphical approaches to identify patterns in datasets; often used to generate hypotheses for testing
3. **Inferential statistics:** making statements about a large group of people or things (a population) using statistics derived from a subgroup (a sample), using probability and sampling theory
4. **Significance testing:** making statements on the observed importance of (inferred) differences

# Terminology

- A **variable** refers to anything that can change or vary e.g rainfall, age, opinions.
- In the social sciences, information about variables is most typically obtained by means of **survey questionnaire**. The answers to the questions are the variables.
- The answers given by different respondents to the same question are called **cases** .
- The data is usually entered into a grid or **data matrix**, with columns formed by the variables and the rows by respondents or cases.
- The **order** in which variables and cases are placed in the grid is **rarely important**.

# Data matrix

|    | A                | B                                         | R                                                                                                                                                                                                                                                   | S                                 | T                    | U                          | V                | W                | X                     | Y                               | Z                        | AA                              | AB               |
|----|------------------|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----------------------|----------------------------|------------------|------------------|-----------------------|---------------------------------|--------------------------|---------------------------------|------------------|
| 1  | Variable Group   |                                           | What does sustainability mean to you? (select all that apply)                                                                                                                                                                                       |                                   |                      |                            |                  |                  |                       |                                 |                          |                                 |                  |
| 2  | Question         | Which gender?                             | Good corporate citizenship                                                                                                                                                                                                                          | Being environmentally responsible | Targeting zero waste | Being socially responsible | Carbon emissions | Using renewables | Protecting ecosystems | Having action on climate change | Having social well-being | Traceability (carbon footprint) | Conserving water |
| 3  | Variable coding: | 1: Male<br>2: Female<br>3: Gender diverse | <p>These numbers don't mean anything <i>per se</i> (nominal data), just that the item was chosen.</p> <p>You can count how many times each item was chosen individually, but analysing combinations within multi-pick questions is problematic!</p> |                                   |                      |                            |                  |                  |                       |                                 |                          |                                 |                  |
| 4  | Variable         | Gender                                    | Corporate citizenship                                                                                                                                                                                                                               | Enviro responsible                | Zero waste           | Social responsible         | Carbon emissions | Renewables       | Ecosystems            | Climate change                  | Social well-being        | Traceability                    | Water            |
| 5  | P1               |                                           |                                                                                                                                                                                                                                                     | 2                                 |                      |                            |                  | 6                |                       |                                 |                          |                                 |                  |
| 6  | P2               |                                           |                                                                                                                                                                                                                                                     | 2                                 | 3                    |                            | 5                | 6                |                       | 8                               |                          | 10                              |                  |
| 7  | P3               |                                           |                                                                                                                                                                                                                                                     | 2                                 |                      |                            |                  |                  | 7                     | 8                               |                          |                                 | 11               |
| 8  | P4               |                                           |                                                                                                                                                                                                                                                     | 2                                 |                      |                            | 5                | 6                |                       |                                 |                          |                                 |                  |
| 9  | P5               |                                           |                                                                                                                                                                                                                                                     | 2                                 | 3                    |                            | 5                | 6                | 7                     | 8                               |                          | 10                              |                  |
| 10 | P6               |                                           |                                                                                                                                                                                                                                                     | 2                                 |                      | 4                          |                  |                  | 7                     |                                 |                          |                                 |                  |
| 11 | P7               |                                           |                                                                                                                                                                                                                                                     | 2                                 |                      | 4                          |                  |                  | 7                     |                                 |                          | 10                              |                  |
| 12 | P8               |                                           |                                                                                                                                                                                                                                                     | 2                                 |                      | 4                          | 5                | 6                | 7                     |                                 |                          |                                 |                  |
| 13 | P9               |                                           |                                                                                                                                                                                                                                                     | 2                                 | 3                    | 4                          | 5                | 6                | 7                     | 8                               | 9                        | 10                              | 11               |
| 14 | P10              |                                           |                                                                                                                                                                                                                                                     | 2                                 | 3                    |                            | 5                | 6                | 7                     |                                 |                          |                                 | 11               |
| 15 | P11              |                                           | 1                                                                                                                                                                                                                                                   | 2                                 |                      | 4                          |                  | 6                |                       |                                 | 9                        | 10                              |                  |
| 16 | P12              |                                           |                                                                                                                                                                                                                                                     | 2                                 | 3                    |                            | 5                | 6                |                       | 8                               | 9                        |                                 |                  |
| 17 | P13              |                                           |                                                                                                                                                                                                                                                     |                                   |                      | 4                          | 5                |                  |                       |                                 |                          | 10                              |                  |
| 18 | P14              |                                           |                                                                                                                                                                                                                                                     | 2                                 |                      |                            |                  |                  |                       | 8                               |                          |                                 |                  |
| 19 | P15              |                                           |                                                                                                                                                                                                                                                     | 2                                 | 3                    | 4                          |                  |                  | 7                     |                                 | 9                        |                                 |                  |
| 20 | P16              |                                           |                                                                                                                                                                                                                                                     | 2                                 |                      | 4                          |                  | 6                | 7                     |                                 | 9                        | 10                              |                  |
| 21 | P17              |                                           | 1                                                                                                                                                                                                                                                   | 2                                 | 3                    | 4                          | 5                | 6                |                       |                                 |                          | 10                              | 11               |
| 22 | P18              |                                           |                                                                                                                                                                                                                                                     | 2                                 | 3                    |                            | 5                | 6                |                       | 8                               |                          |                                 |                  |

# Individual vs Grouped Data

- **Individual data** occurs when each data value relates to an individual.  
e.g: **Age** (21 vs 59)
- **Grouped data** occurs when individual values have been grouped into specific, pre-determined classes (should be *mutually exclusive*) [also called *categories* or *classes*]  
e.g: **Age bands** (20-24 vs 55-59)

# Discrete vs Continuous Data

- **Discrete Data:** generally whole numbers, a count of values in a given range

e.g: Age (21 vs 59; 12 × 20-24s)

- **Continuous Data:** potentially infinite number of values within a range – measured precisely and certainly to decimal places (in theory)

e.g: time it takes a customer to be greeted in store

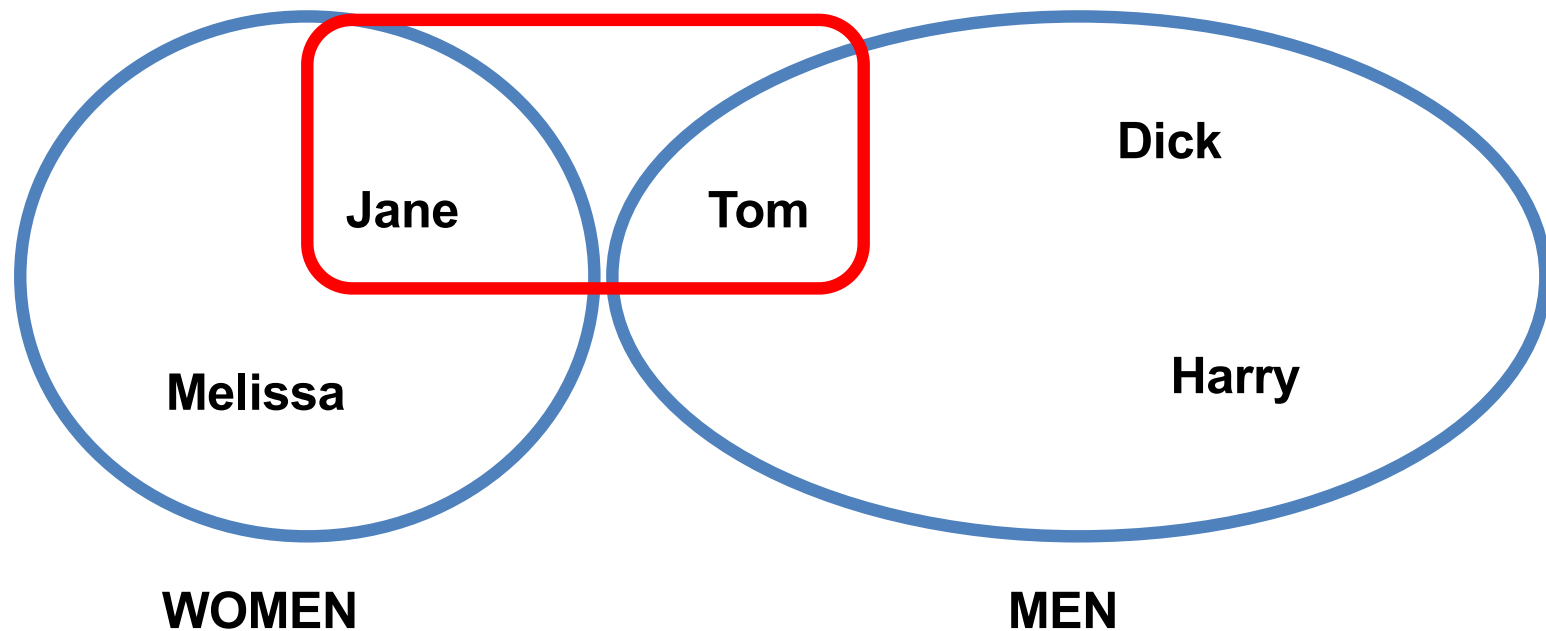
# Levels of Measurement

- Nominal
  - Simple Categories
- Ordinal
  - Hierarchy / ranking
- Interval
  - Has order and equal intervals
- Ratio/Scale
  - Has a ‘meaningful zero’

# Nominal data

- The nominal is the most primitive level of measurement since it simply involves placing objects into categories or classifications
- Each category is homogeneous and **mutually exclusive**.
- Numbers or symbols are assigned to the categories so that each category has a unique identifier.
- The categorisation should be **exhaustive** so that all objects can be assigned.
- Binary coding of data is a special case of the nominal scale where only two categories recognised (e.g gender; yes/no)
- Because of lack of order or equal intervals, one cannot perform arithmetic (+, -, /, \*) or logical operations (>, <, =) on nominal data. Can perform frequency counts & calculate %

# Runners in a road race



|                 | MEN (A) | WOMEN (B) |
|-----------------|---------|-----------|
| RED SHIRTS (1)  |         |           |
| BLUE SHIRTS (2) |         |           |



# Ordinal data

- Ranking of categories: Largest to smallest, best to worst
- Ordinal data is produced when a set of objects is arranged in order of magnitude but the exact differences between successive objects have not been calculated.
- In ordinal data it can be concluded that A is bigger than B but not by how much.
- Intervals vary in size so the largest may be much bigger than the next largest who may be only slightly bigger than the third largest.
- Arithmetic operations are meaningless with ordinal data, which are restricted to logical operations (more than, less than, equal to).

For instance, the fifth-ranked pupil in a school class is not twice as clever as the pupil ranked tenth.

# Ordinal data

| <i>Finishing order in road race</i> |            |              |
|-------------------------------------|------------|--------------|
|                                     | <b>MEN</b> | <b>WOMEN</b> |
| 1                                   | Tom        | Jane         |
| 2                                   | Dick       | Melissa      |
| 3                                   | Harry      | Laura        |

Ordinal measures may be

- complete orderings eg road race 1, 2, 3, etc
- partial orderings if includes ties eg goals scored in last game
- ranked categories eg questionnaire items “How often do you go swimming” (Daily, Weekly, Monthly, less frequently, never)

# Interval data

- Interval data are always numeric but could be continuous or discrete (counts)
- Interval data has order and equal intervals.
- Intervals of the same size have the same meaning. e.g the difference between 20°C and 25°C ( $25 - 20 = 5^\circ\text{C}$ ) is the same as the difference between 7°C and 12°C ( $12 - 7 = 5^\circ\text{C}$ ).  
e.g the difference between -5°F and 5°F ( $5 - (-5) = 10^\circ\text{F}$ ) is the same as the difference between 100°F and 110°F.
- Interval scales have an arbitrary zero point
- With interval data, addition and subtraction are possible but not multiplication or division

# Interval data

- **Arrival time at the finish of the road race**

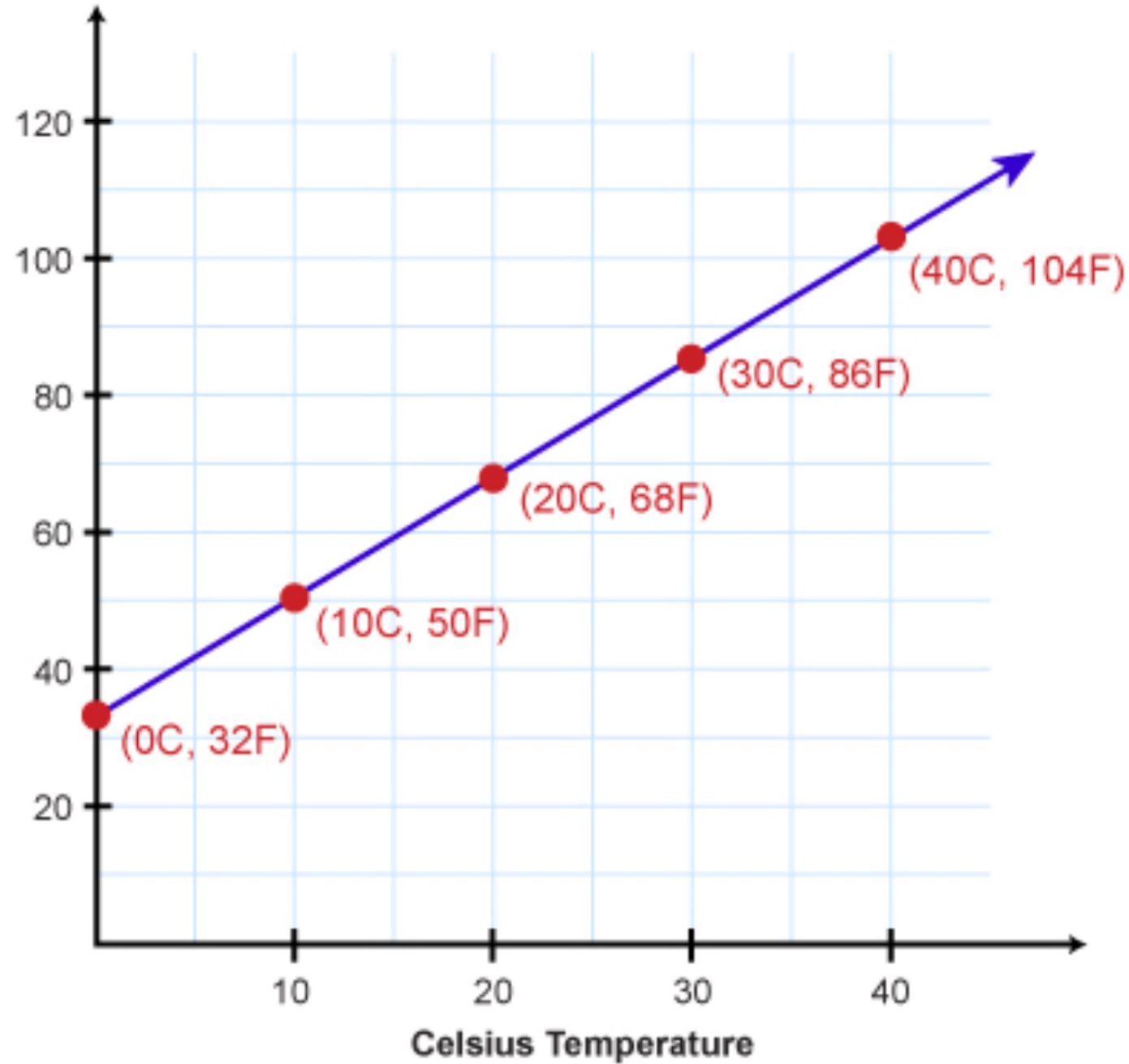


- The times do not have a fixed zero start point
- Men and women racers could have started at different times.

# Ratio data

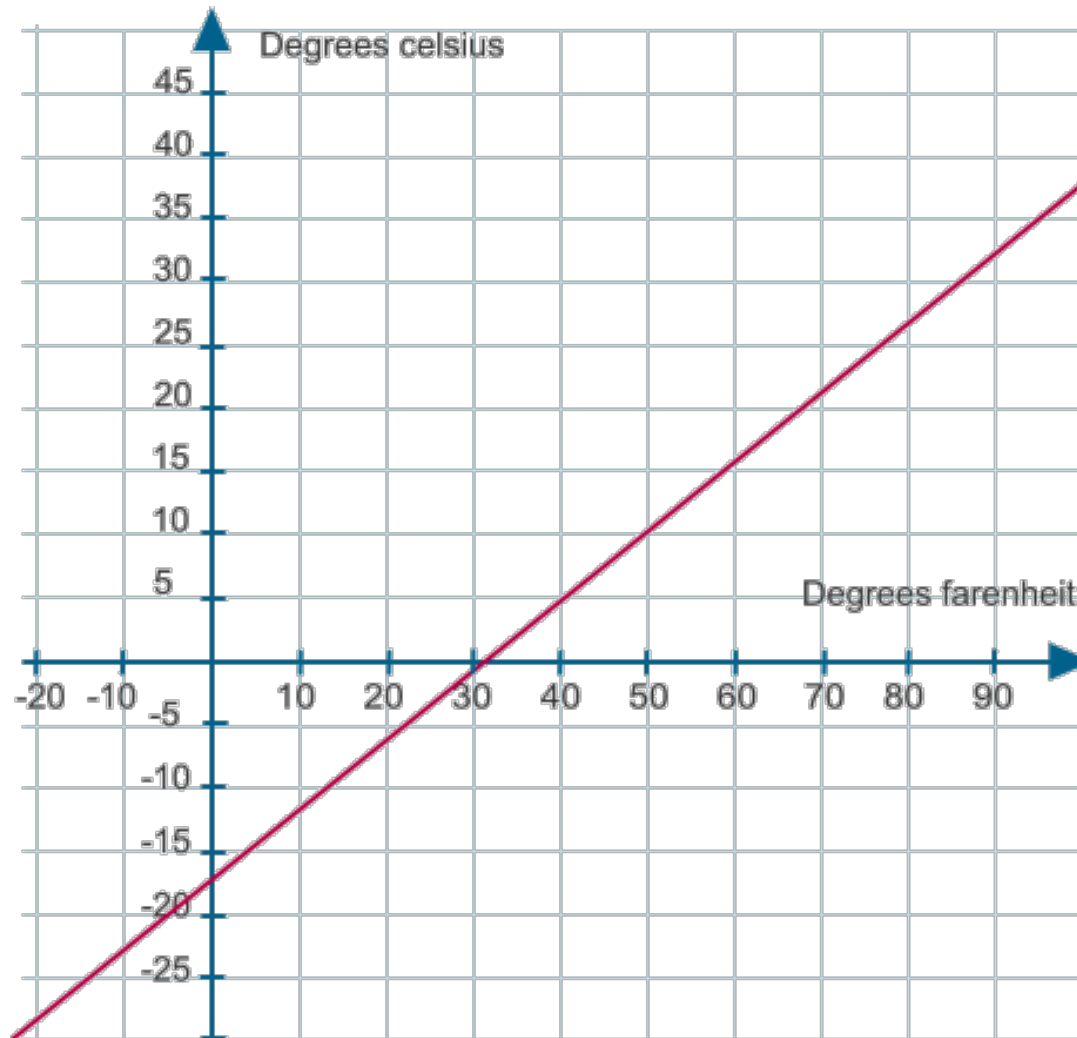
- Ratio data possesses all the properties of an interval scale, including order and equal intervals
- Ratio scales have an absolute zero point (the 0 is “meaningful”).
- With ratio data division and multiplication are possible
- Thus \$20 is twice as much as \$10 and that ratio holds true no matter what currency is used.
- A ratio measure is independent of the scale on which it is based. Ratio of heights 165cm to 175cm is 0.94. Same ratio in inches: 412.5 to 437.5 is 0.94.
- Try weight: if 1lb = 453g, what does 2lbs equate to?
- Try temperature: if 10°C is 50°F, what does 20°C equal?

# Conversion of Celsius Temperatures to Fahrenheit



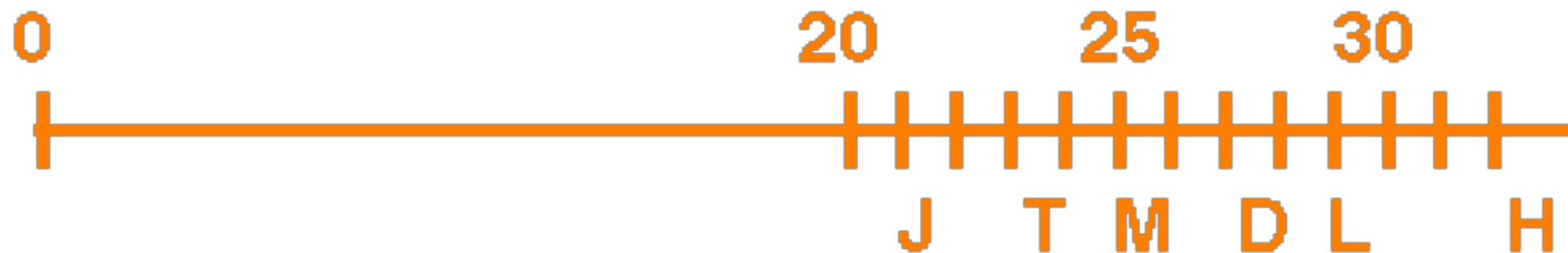
# Celsius vs Fahrenheit

$$F = \frac{C \times 5}{9} + 32 \quad C = \frac{(F-32) \times 5}{9}$$



# Ratio data

## Elapsed running time



Ratio measures have:

- a fixed zero that means 'no quantity';
- a constant interval (distance on the scale).

# Qualitative-Quantitative Continuum

Qualitative

Quantitative



text   nominal

ordinal

interval & ratio

# Chart Choices

***NOMINAL***

***or***

***ORDINAL***

***Bar graph***

***Pie graph***

***Stacked bar***

***INTERVAL***

***or***

***RATIO***

***Histogram***

***or***

***Scatterplot***

# Central Tendency: the ‘average’

- It is more convenient to describe a set of numbers by using a *single number*
- Calculating a single *descriptive* number is one of the most frequently encountered methods of summarising (condensing) data
- The *average* is simply any single figure that is representative of many numbers
- The term is also used to mean an average calculated as the sum of a set of numbers divided by how many numbers there are
- The term “*measure of central tendency*” describes the idea of a *typical value*, and the term *mean* is used for the specific average described above
- But there are others!

$$\bar{x} = \frac{\sum x}{n}$$

X-bar      Sigma

# Central Tendency

- “Average values”
- **Mode**
  - Largest frequency count
  - Nominal/grouped data, ordinal, interval, ratio data
  - Modality: bimodal
- **Median**
  - ‘Middle value’
  - Ordinal, Interval, Ratio data
  - Good for extreme or skewed data (e.g income, house values)
- **Mean**
  - Interval/ratio data
  - $\bar{x}$  (“x-bar”) for samples,  $\mu$  (“mu”) for population
  - ‘Trimmed mean’: outliers
  - Arithmetic vs Geometric mean (nb: geometric mean not used in this course: good for serially correlated data)

# Measures of Central Tendency

- $n$  = number of observations/cases
- $i$  denotes a particular case
- Mode: case that occurs most frequently
- Median: middle case  $= \frac{n+1}{2}$

# Measures of Central Tendency: Mean

- Summation notation:  $\Sigma$
- $n$  = number of observations/cases
- $i$  denotes a particular case

- Mean  $\bar{x} = \frac{\sum x_i}{n} \quad \bar{x} = \frac{x_1 + x_2 + x_3 + x_4 + x_n}{n}$

notation:  $\sum_{i=1}^n x_i$

# Arithmetic vs Geometric Mean

FYI only!  
(not tested)

- If your portfolio returns each year were:  
**90%, 10%, 20%, 30%, and -90%**  
what would your average return be during  
this period?

- Arithmetic mean:  $\bar{x} = \frac{\sum x_i}{n}$   $\bar{x} = \sum \frac{0.9 + 0.1 + 0.2 + 0.3 - 0.9}{5}$   
 $\bar{x} = 0.12$  (12%) *Excel: =average()*
- Geometric mean:  $GM = [(1 + x_1)(1 + x_2)(1 + x_3) \dots (1 + x_n)]^{\frac{1}{n}} - 1$  *Excel: =geomean()*  
 $GM = [(1 + 0.9)(1 + 0.1)(1 + 0.2)(1 + 0.3)(1 - 0.9)]^{\frac{1}{5}} - 1$   
 $GM = [0.326]^{\frac{1}{5}} - 1 = -0.201$  (-20%)

# Measures of Central Tendency

Mean = 24.93

$$33 + 31 + 7 + 34 + 16 + 34 + 6 + 25 \\ + 37 + 17 + 32 + 11 + 41 + 20 + 30$$

$$\bar{x} = \frac{374}{15} = 24.9$$

Median = 30

$$6 + 7 + 11 + 16 + 17 + 20 + 25 \\ + 30 +$$

Mode = 34

$$31 + 32 + 33 + \underline{34 + 34} + 37 + 41$$

# Measures of Central Tendency

- ~~Mean = 24.93~~
- Mean = 23.75

$$33 + 31 + 7 + 34 + 16 + 34 + 6 + 25 \\ + 37 + 17 + 32 + 11 + 41 + 20 + 30 \\ + 6$$

~~$$\bar{x} = \frac{374}{15} = 24.9$$~~

- ~~Median = 30~~
- Median = 27.5

$$\bar{x} = \frac{380}{16} = 23.75$$

- ~~Mode = 34~~
- Mode = 6 & 34 (bimodal)

$$\underline{6} + \underline{6} + 7 + 11 + 16 + 17 + 20 \\ + \boxed{25 + 30} + \\ 31 + 32 + 33 + \underline{34 + 34} + 37 + 41$$

# Measures of Central Tendency

$$\begin{aligned}
 &6 + 6 + 7 + 11 + 16 + 17 + 20 \\
 &\quad + 25 + 30 + \\
 &31 + 32 + 33 + 34 + 34 + 37 + 41
 \end{aligned}$$

$$N = 16 \quad \bar{x} = 23.75$$

| $x_i$     |       | $x_i$     |       |
|-----------|-------|-----------|-------|
|           |       | 6         |       |
| 6         |       | 6         |       |
| 7         |       | 7         |       |
| 11        |       | 11        |       |
| 16        |       | 16        |       |
| 17        |       | 17        |       |
| 20        |       | 20        |       |
| 25        |       | 25        |       |
| 30        |       | 30        |       |
| 31        |       | 31        |       |
| 32        |       | 32        |       |
| 33        |       | 33        |       |
| 34        |       | 34        |       |
| 34        |       | 34        |       |
| 37        |       | 37        |       |
| 41        |       | 41        |       |
| <hr/>     |       | <hr/>     |       |
| N         | 15    | N         | 16    |
| $\Sigma$  | 374   | $\Sigma$  | 380   |
| $\bar{x}$ | 24.93 | $\bar{x}$ | 23.75 |

→

$$\bar{x} = \frac{\sum x_i}{n}$$

# Frequency Data

## Deriving $\bar{X}$ manually:

| X         | f   | f x  |
|-----------|-----|------|
| 1         | 33  | 33   |
| 2         | 53  | 106  |
| 3         | 240 | 720  |
| 4         | 261 | 1044 |
| 5         | 154 | 770  |
| 6         | 301 |      |
| N / Sum   |     |      |
| $\bar{X}$ |     |      |

$$\bar{x} = \frac{\sum fx}{\sum f}$$

How often do you buy a product that has come from a business that implements sustainable practices?

| AY            |
|---------------|
| Sustainable   |
| How often do  |
| 1: Not at all |
| 5: Very often |
| 6: Don't know |

where:  $\bar{x}$  = the mean of the distribution  
 $f$  = the frequency count

- A straight 'average' (eg arithmetic mean) treats all observations with equal weight

Eg marks for assessments in the paper:

70, 80, 90, 60     $\bar{x} = 75$

## Weighted Averages

- A *weighted* average applies different levels of importance to observations

Eg the assessments are worth 10%, 20%, 10%, 60%

=  $70 \times 0.1 + 80 \times 0.2 + 90 \times 0.1 + 60 \times 0.6$      $\bar{x} = 68$

| Order | Units Ordered | Cost per unit \$ | Total Cost of order |
|-------|---------------|------------------|---------------------|
| 1     | 50            | 240.00           | \$12,000.00         |
| 2     | 100           | 200.00           | \$20,000.00         |
| 3     | 100           | 220.00           | \$22,000.00         |
| 4     | 50            | 280.00           | \$14,000.00         |
| Total | 300           | <b>226.67</b>    | \$68,000.00         |

Weighted  
Average  
**COGS**

# Grouped Data

- Use midpoint method

$$\text{Mid} = \left( \frac{(\text{top of interval} - \text{bottom of interval})}{2} \right) + \text{bottom of interval}$$

$$\text{Mid} = \left( \frac{(40 - 30)}{2} \right) + 30 = 5 + 30 = 35$$

# Grouped Data

- Sort Raw Data in Ascending Order (journey to work times in mins)  
12, 13, 17, 21, 24, 24, 26, 27, 27, 30, 32, 35, 37, 38, 41, 43, 44, 46, 53, 58
- Find Range:  $58 - 12 = 46$
- Select Number of Classes: **5 (usually between 5 and 15)**
- Compute Class Interval (Width): **10** ( $46/5$  then round up)
- Determine Class Boundaries (Limits): **10, 20, 30, 40, 50, 60**
- Compute Class Midpoints
- Count Observations & Assign to Classes

# Grouped Data

Raw Data: (journey to work times in mins) **12, 13, 17, 21, 24, 24, 26, 27, 27, 30, 32, 35, 37, 38, 41, 43, 44, 46, 53, 58**

| Stated class intervals (mins) | Midpoints           |
|-------------------------------|---------------------|
| 10-19                         | $10 + 0.5(10) = 15$ |
| 20-29                         | $20 + 0.5(10) = 25$ |
| 30-39                         | 35                  |
| 40-49                         | 45                  |
| 50-59                         | 55                  |

**Midpoint** = Lower limit + half of the interval length

# Grouped Data

Class intervals and midpoints  
(rounded values)

| Stated class intervals (mins) | Midpoints           |
|-------------------------------|---------------------|
| 10-19                         | $10 + 0.5(10) = 15$ |
| 20-29                         | $20 + 0.5(10) = 25$ |
| 30-39                         | 35                  |
| 40-49                         | 45                  |
| 50-59                         | 55                  |

**Midpoint** = Lower limit + half of the interval length

# Grouped Data

## Data in Ordered Array:

12, 13, 17, 21, 24, 24, 26, 27, 27, 30, 32, 35, 37, 38, 41, 43, 44, 46, 53, 58

| Class   | mid | Frequency | Relative Frequency | %   |
|---------|-----|-----------|--------------------|-----|
| 10 – 20 | 15  | 3         | 0.15               | 15  |
| 20 – 30 | 25  | 6         | 0.30               | 30  |
| 30 – 40 | 35  | 5         | 0.25               | 25  |
| 40 – 50 | 45  | 4         | 0.20               | 20  |
| 50 – 60 | 55  | 2         | 0.10               | 10  |
| Total   |     | 20        | 1.00               | 100 |
|         |     |           |                    |     |

# Grouped Data

## Data in Ordered Array:

12, 13, 17, 21, 24, 24, 26, 27, 27, 30, 32, 35, 37, 38, 41, 43, 44, 46, 53, 58

| Class   | mid |         | Frequency | Relative Frequency | %   | weight |
|---------|-----|---------|-----------|--------------------|-----|--------|
| 10 – 20 | 15  | =15 x 3 | 3         | 0.15               | 15  | 45     |
| 20 – 30 | 25  |         | 6         | 0.30               | 30  | 150    |
| 30 – 40 | 35  |         | 5         | 0.25               | 25  |        |
| 40 – 50 | 45  |         | 4         | 0.20               | 20  | 180    |
| 50 – 60 | 55  |         | 2         | 0.10               | 10  |        |
| Total   |     |         | 20        | 1.00               | 100 |        |
| Mean    |     |         |           |                    |     |        |
| Median  |     |         |           |                    |     |        |

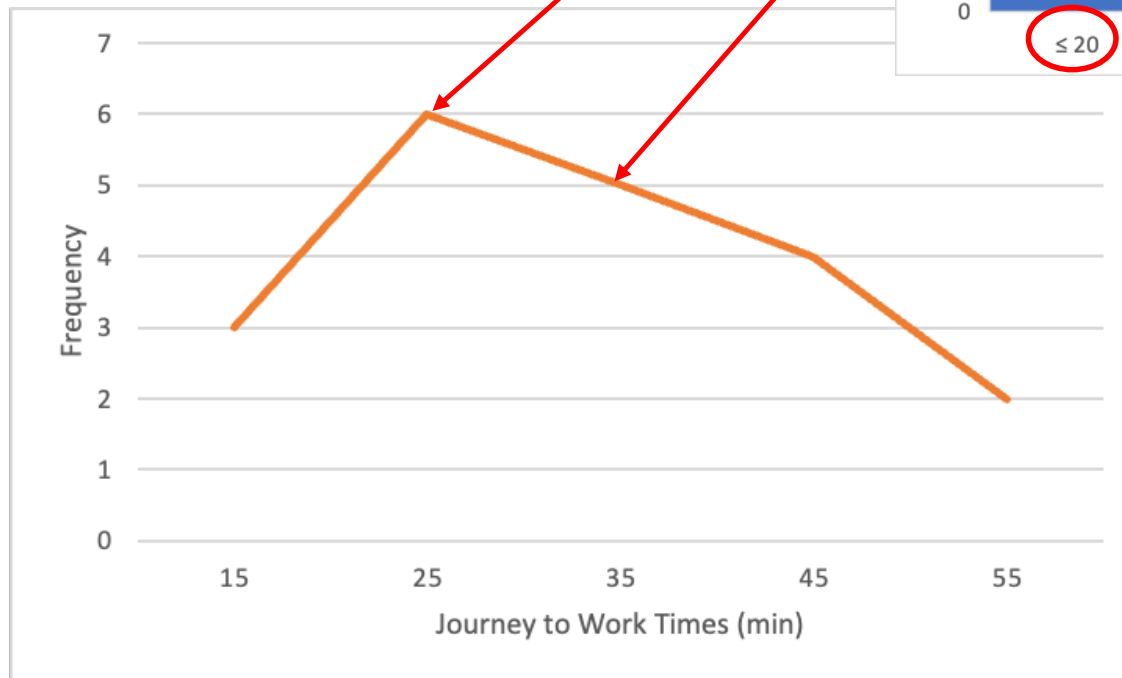
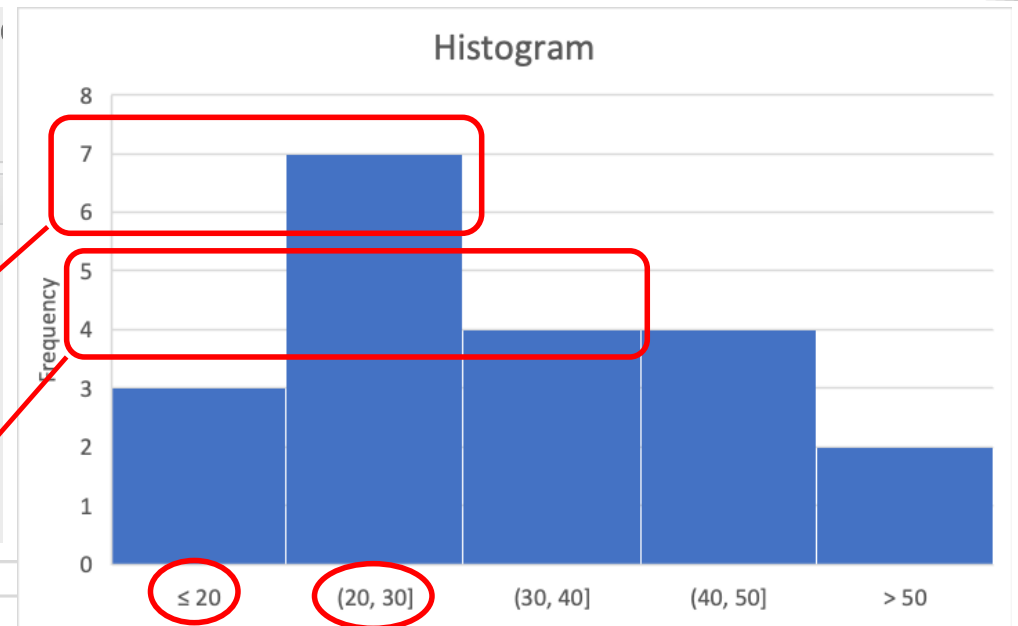
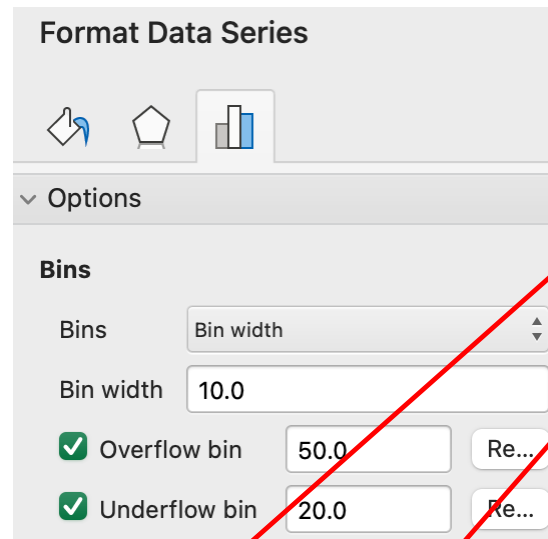
# Grouped Data

## Data in Ordered Array:

12, 13, 17, 21, 24, 24, 26, 27, 27, 30, 32, 35, 37, 38, 41, 43, 44, 46, 53, 58

| Class   | mid | Frequency | Relative Frequency | % weighted |     |
|---------|-----|-----------|--------------------|------------|-----|
| 10 – 20 | 15  | 3         | 0.15               | 15         | 45  |
| 20 – 30 | 25  | 6         | 0.30               | 30         | 150 |
| 30 – 40 | 35  | 5         | 0.25               | 25         | 175 |
| 40 – 50 | 45  | 4         | 0.20               | 20         | 180 |
| 50 – 60 | 55  | 2         | 0.10               | 10         | 110 |
| Total   |     | 20        | 1.00               | 100        | 660 |
| Mean    |     |           |                    | 33         |     |
| Median  |     |           |                    | 30-40      |     |

# Grouped Data: Charts

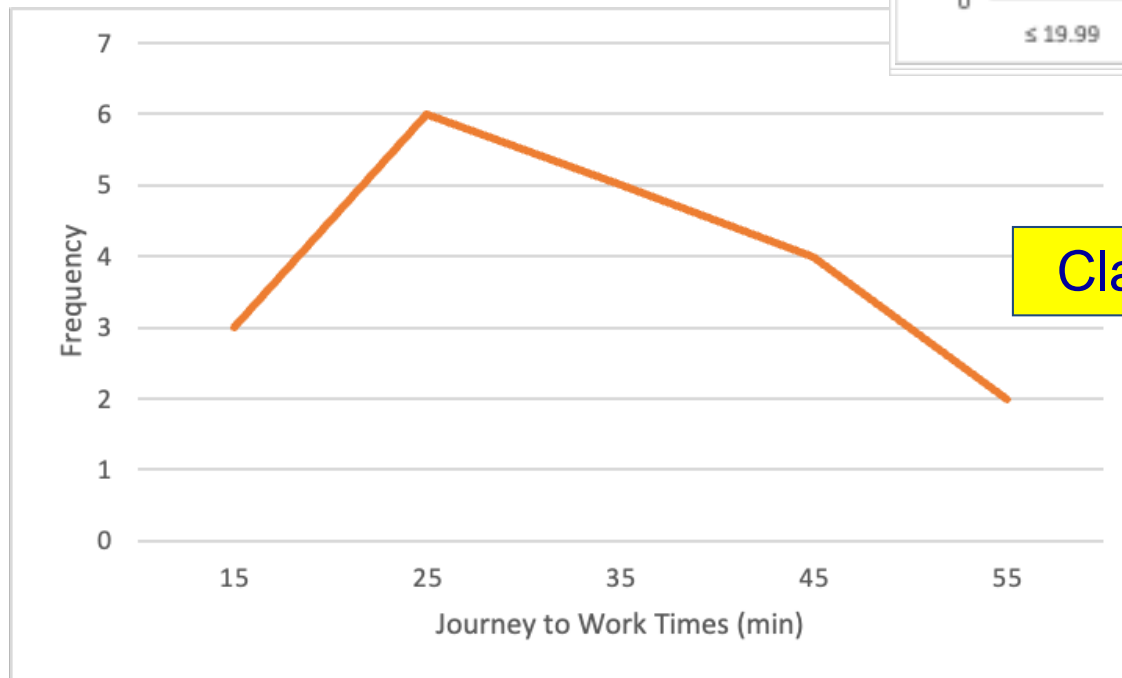
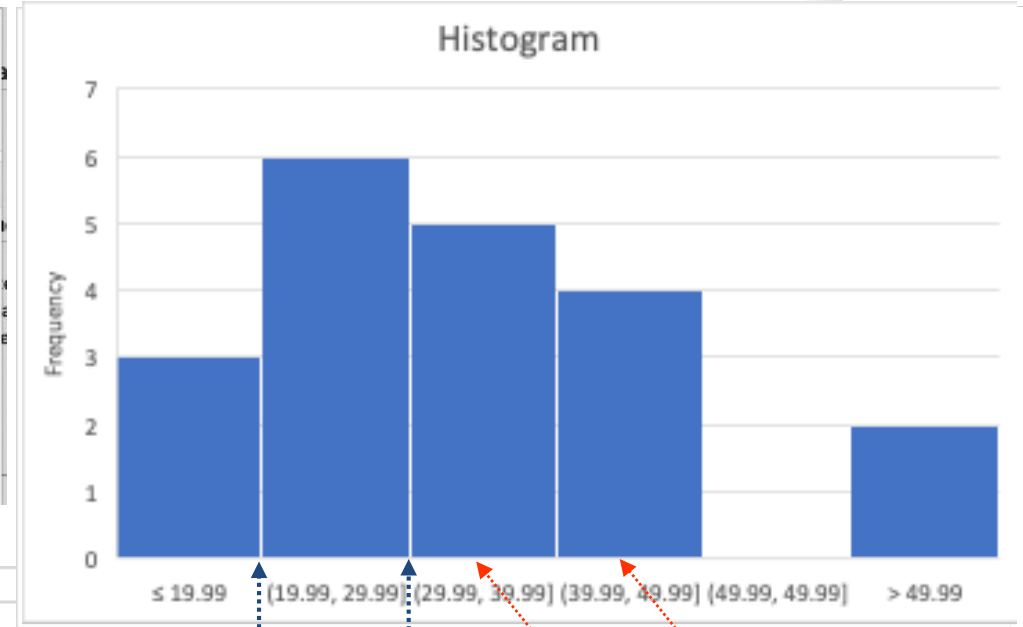
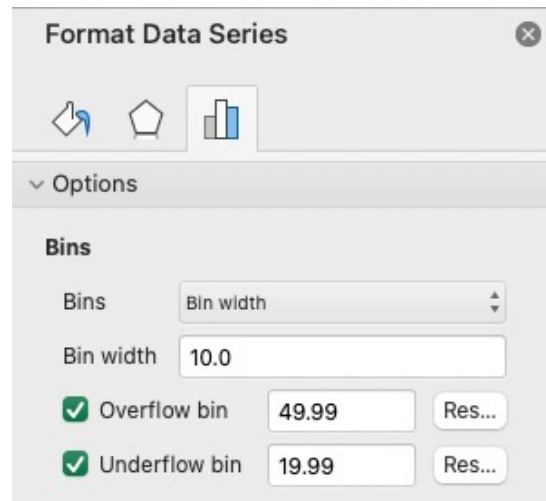


Excel sets the upper class boundary to be  $\leq$   
so 30 falls into the 20-30 class  
Very difficult to adjust in Excel  
where the class boundaries fall

## Data in Ordered Array:

12, 13, 17, 21, 24, 24, 26, 27, 27, 30,  
32, 35, 37, 38, 41, 43, 44, 46, 53, 58

# Grouped Data: Charts



**Class Boundaries**

**Class Midpoints**

**Data in Ordered Array:**

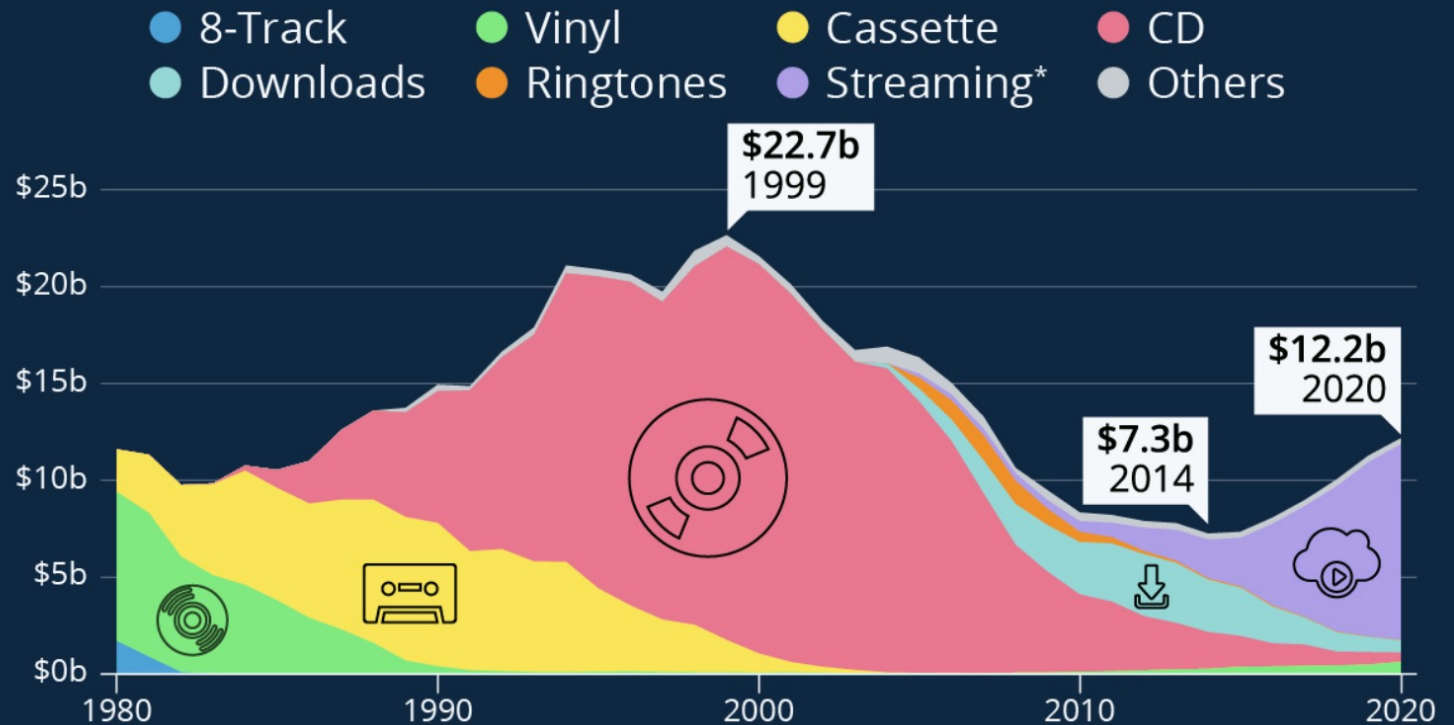
12, 13, 17, 21, 24, 24, 26, 27, 27, 30.1,  
32, 35, 37, 38, 41, 43, 44, 46, 53, 58

# US Music Industry

<https://www.statista.com/chart/17244/us-music-revenue-by-format/>

## From Tape Deck to Tidal: 40 Years of U.S. Music Sales

U.S. recorded music revenues from 1980-2020,  
by format (adjusted for inflation)



\* incl. SoundExchange payments to performers and copyright holders  
for digital and customized radio services

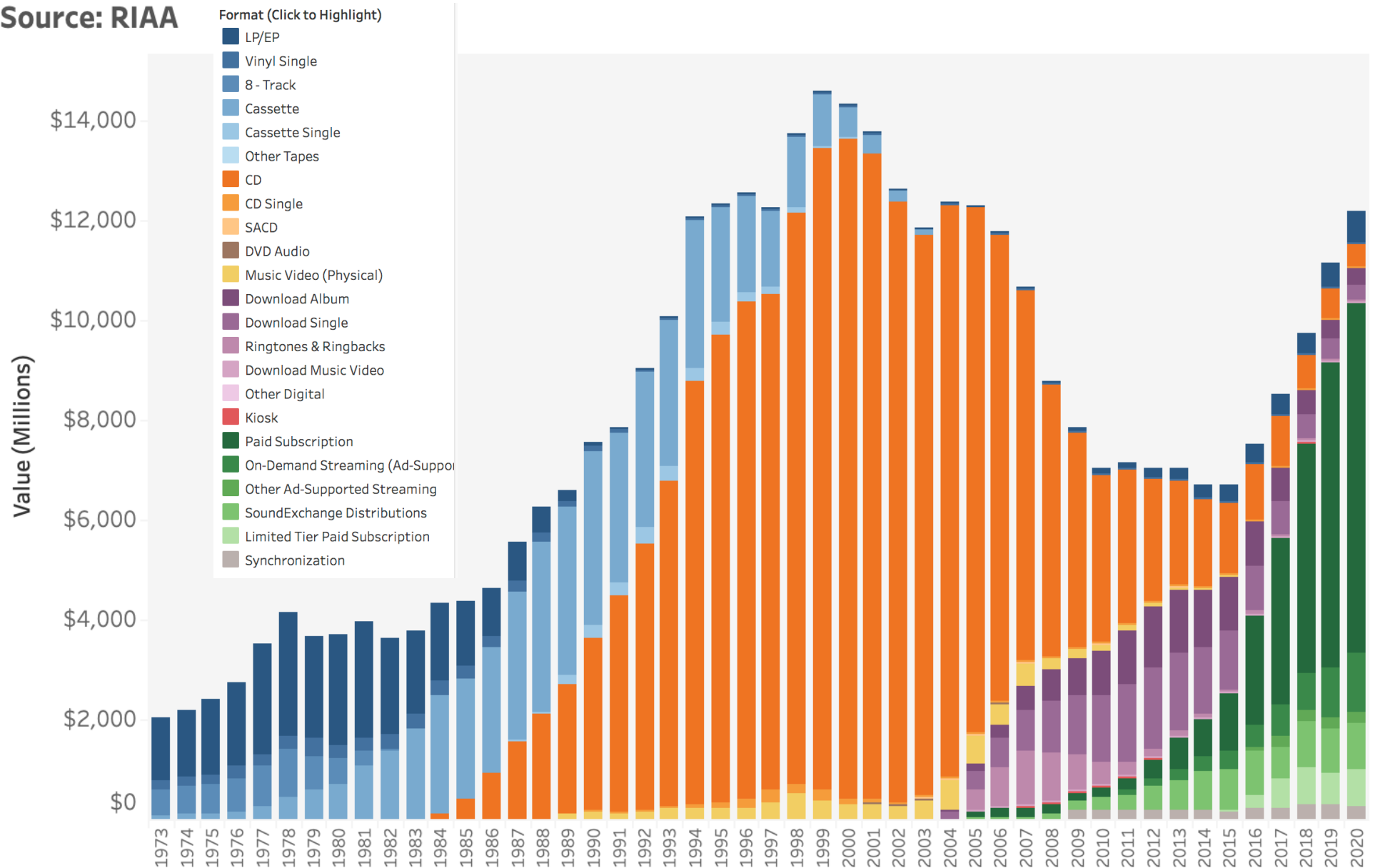
Source: RIAA



# US Music Industry

1973 to 2020, Format(s): All

Source: RIAA

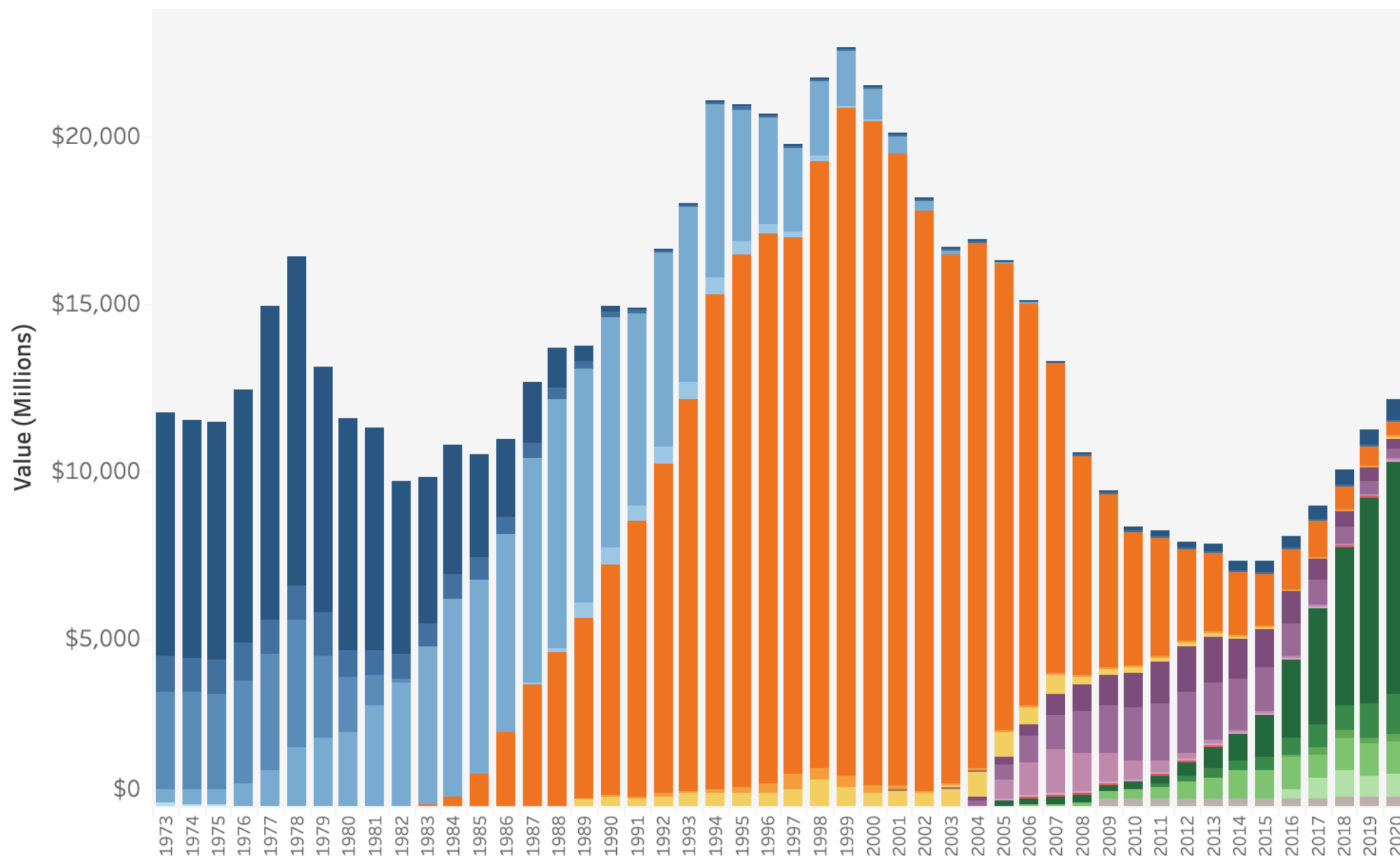


# U.S. Recorded Music Revenues by Format

Adjusted for Inflation, 2020 Dollars

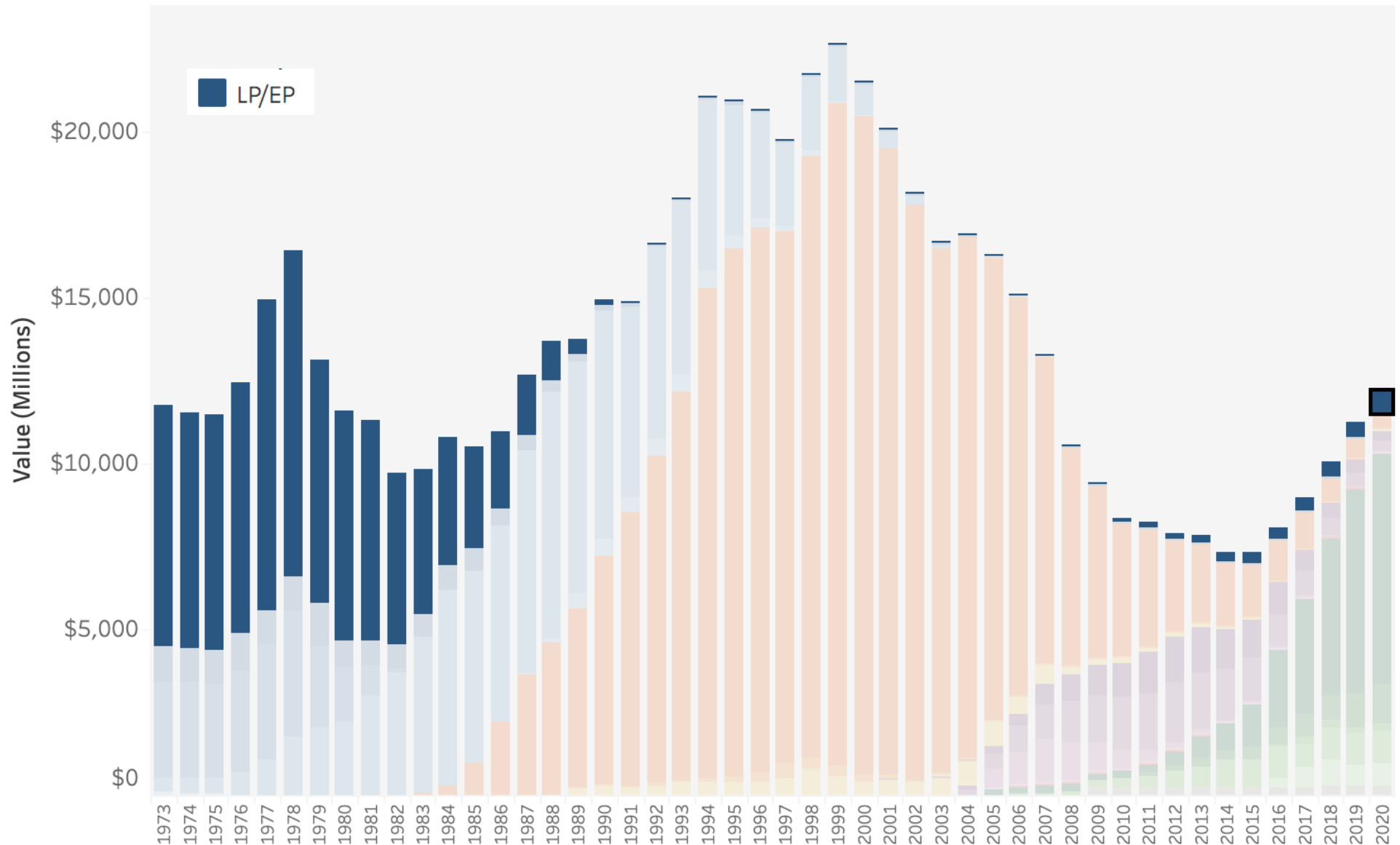
1973 to 2020, Format(s): All

Source: RIAA



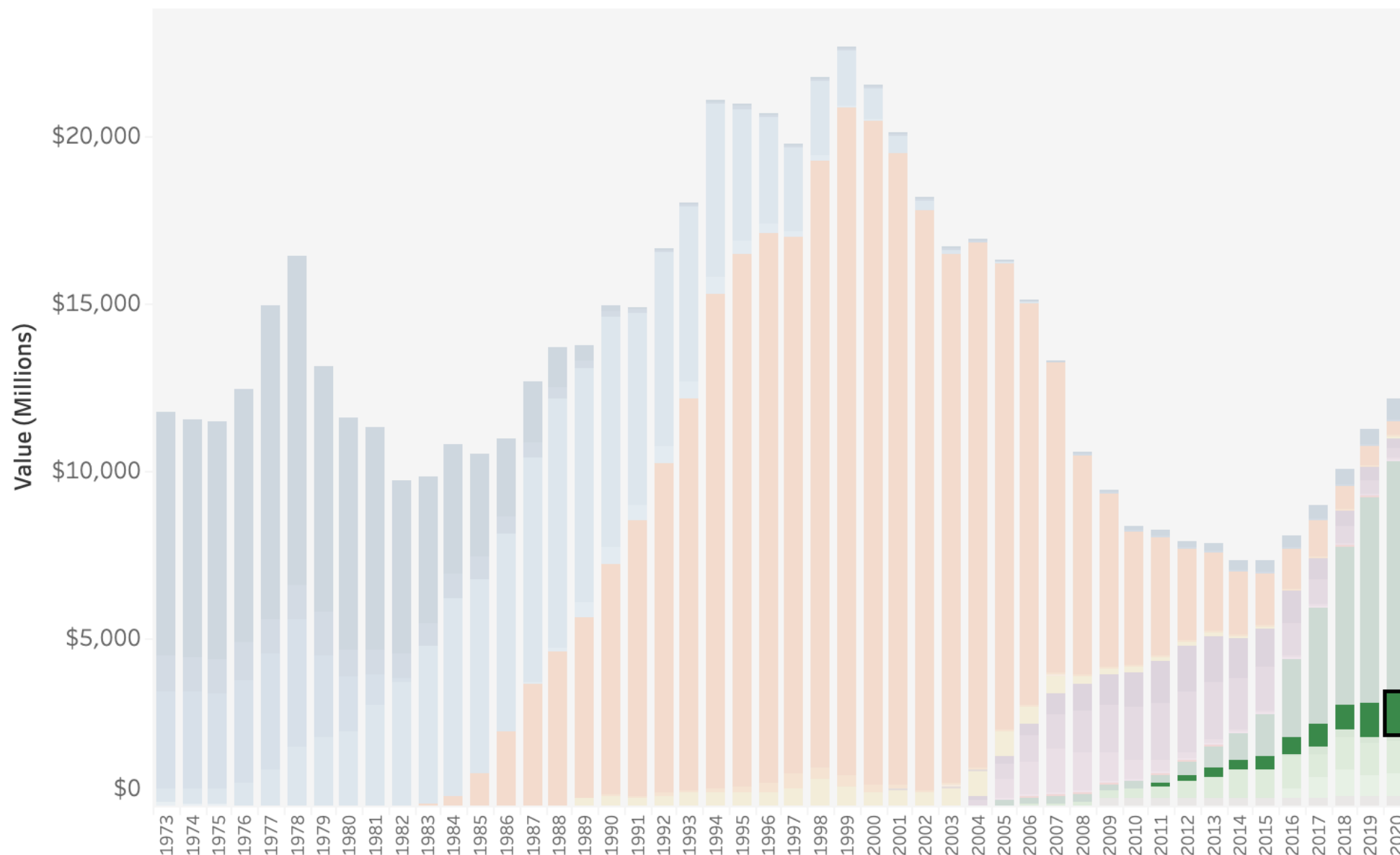
# US Music Industry

**2020:** Revenues for the **LP/EP** format were **\$619.6M** and accounted for **5.1%** of total revenue of **\$12.2B** for all selected formats for the year (adjusted for inflation, 2020 dollars).



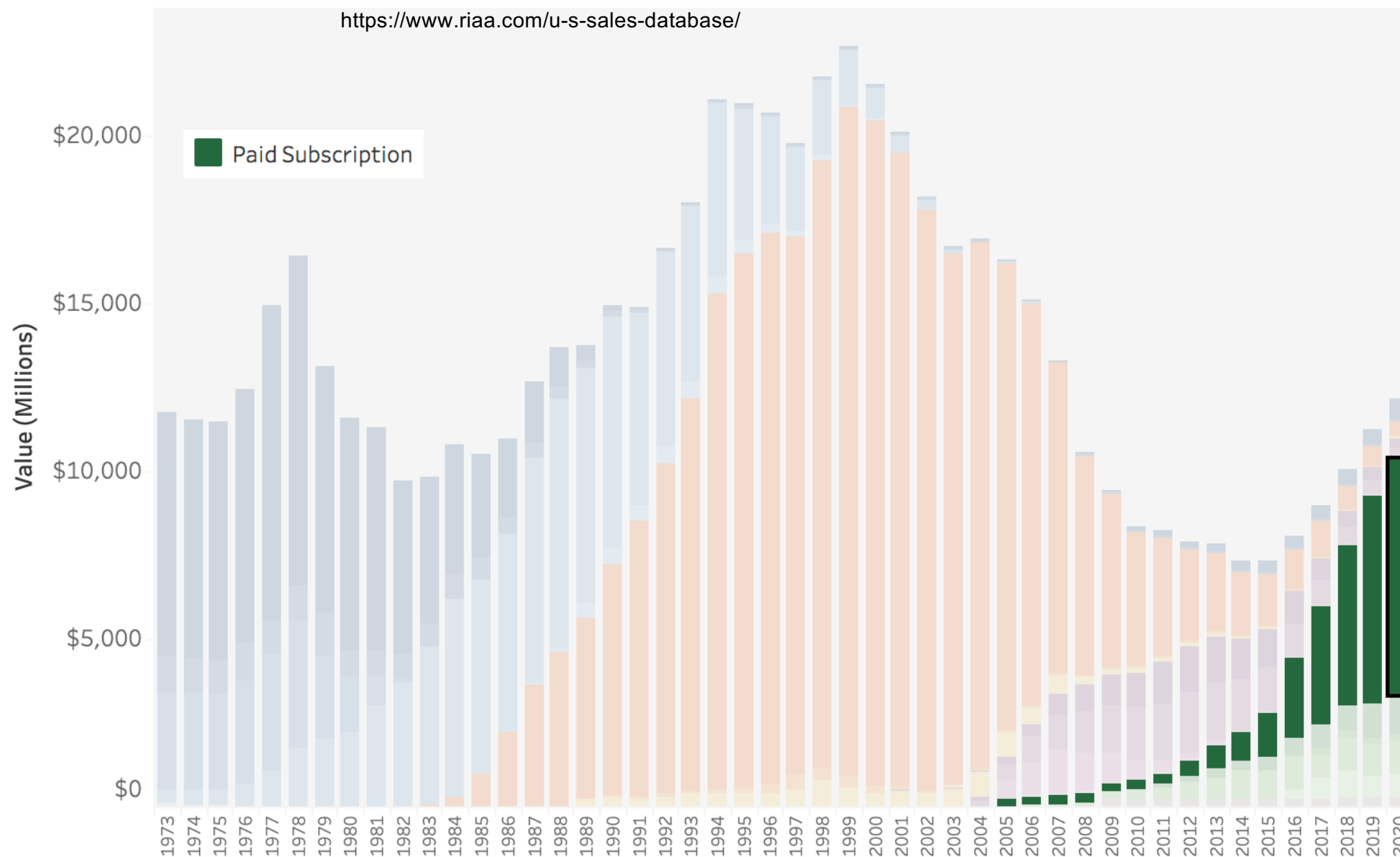
# US Music Industry

**2020:** Revenues for the **On-Demand Streaming (Ad-Supported)** format were **\$1.2B** and accounted for **9.7%** of total revenue of **\$12.2B** for all selected formats for the year (adjusted for inflation, 2020 dollars).



# US Music Industry

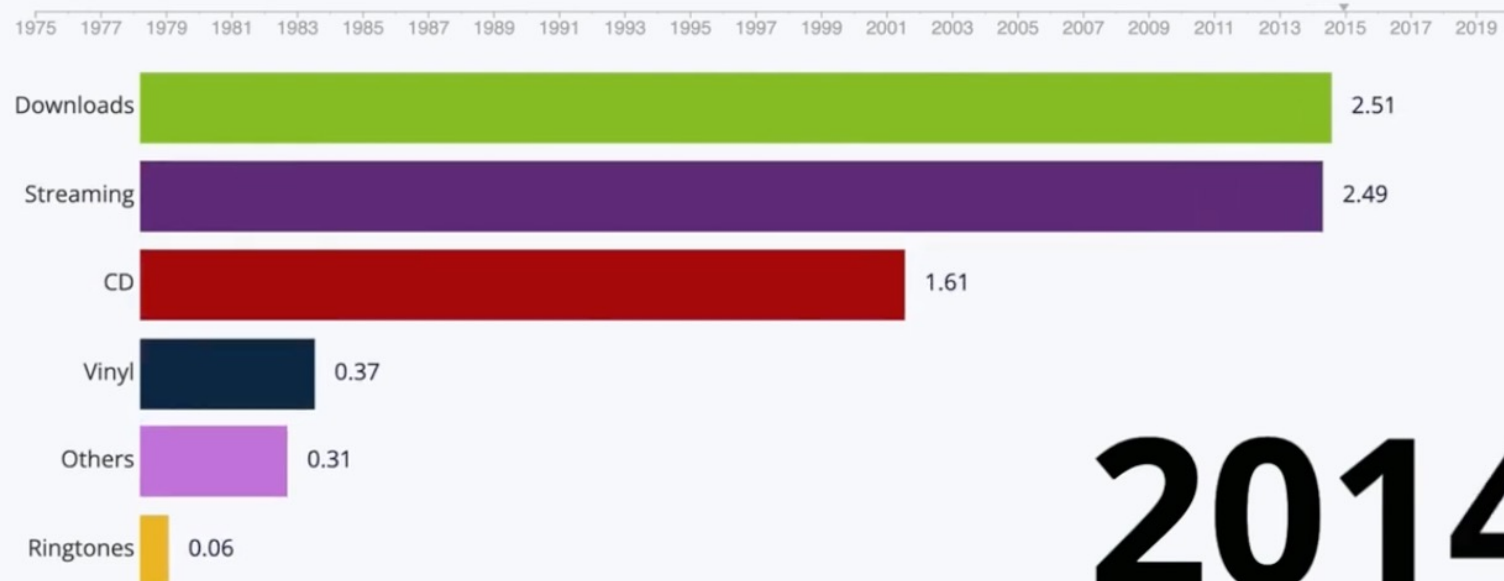
**2020:** Revenues for the **Paid Subscription** format were **\$7.0B** and accounted for **57.7%** of total revenue of **\$12.2B** for all selected formats for the year (adjusted for inflation, 2020 dollars).



# US Music Industry Revenues 1975-2020

## From Tape to Tidal

U.S. music industry revenues from 1975-2020, by format (in billion 2020 U.S. dollars)



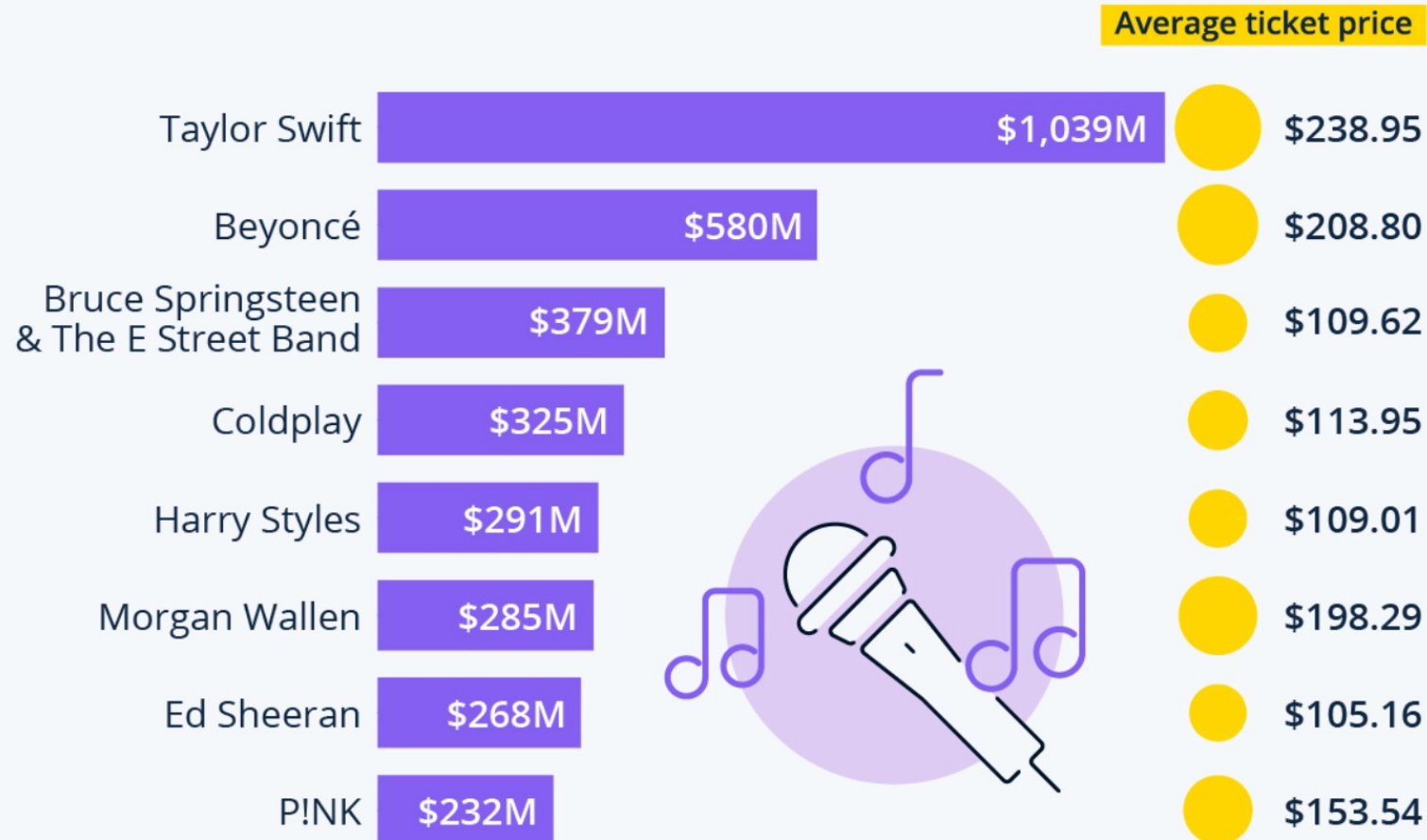
Source: RIAA

**2014**  
statista

<https://www.statista.com/topics/4948/music-industry/#topicOverview>

# The World's Highest-Grossing Tours of 2023

Box office gross of the highest-grossing music tours worldwide in 2023



Source: Pollstar



# Most successful music tours worldwide 2025 (average gross revenue, US\$m)

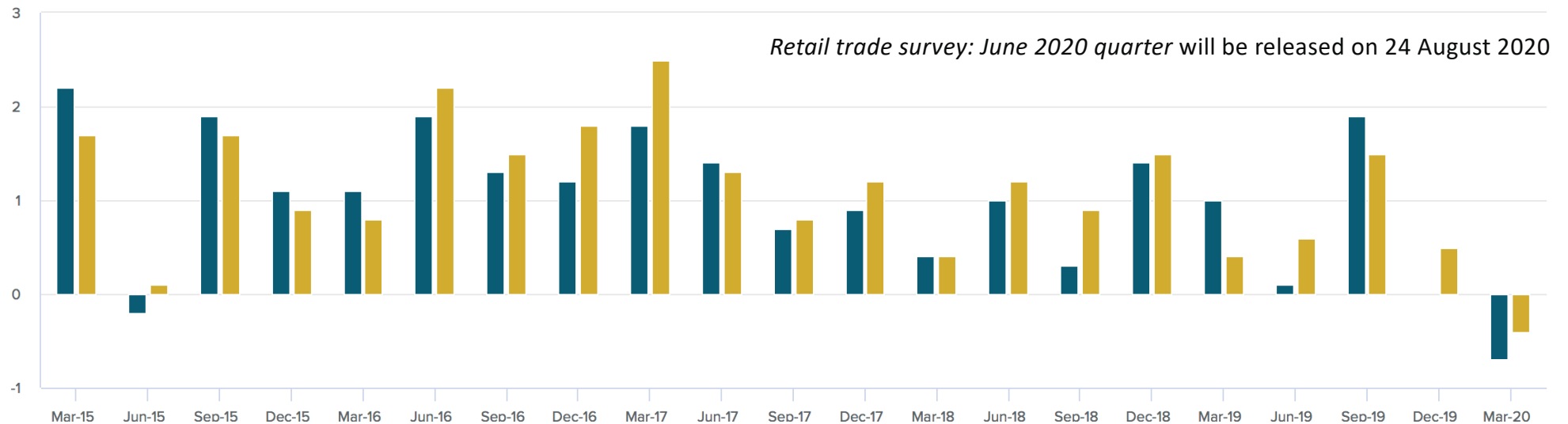


# NZ Retail Survey: Seasonal Adjust

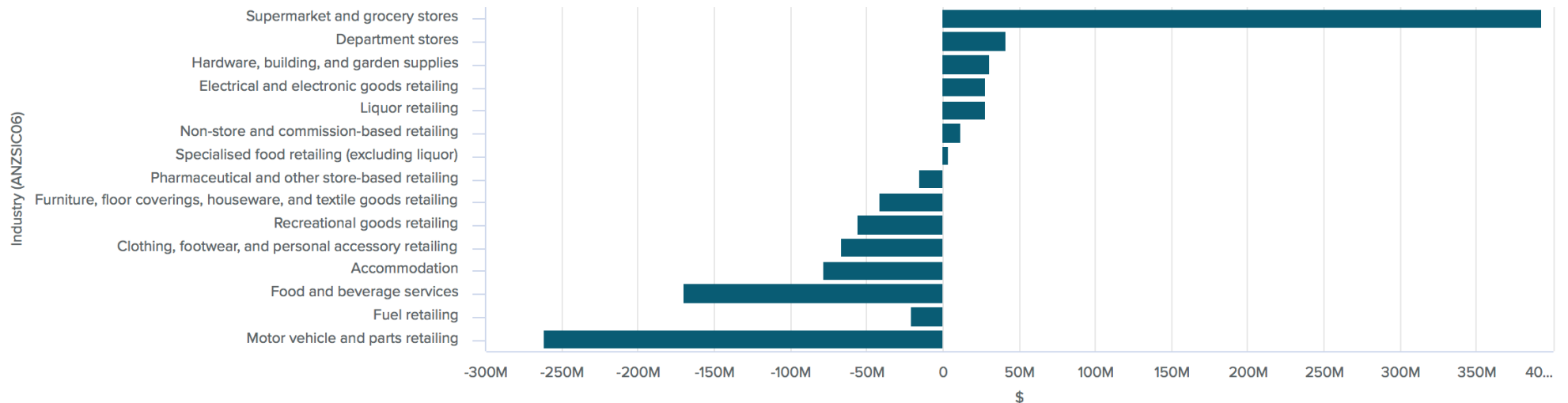
## March 2015 to March 2020

<https://www.stats.govt.nz/information-releases/retail-trade-survey-march-2020-quarter>

Percentage change in seasonally adjusted retail sales, quarterly, March 2015–March 2020



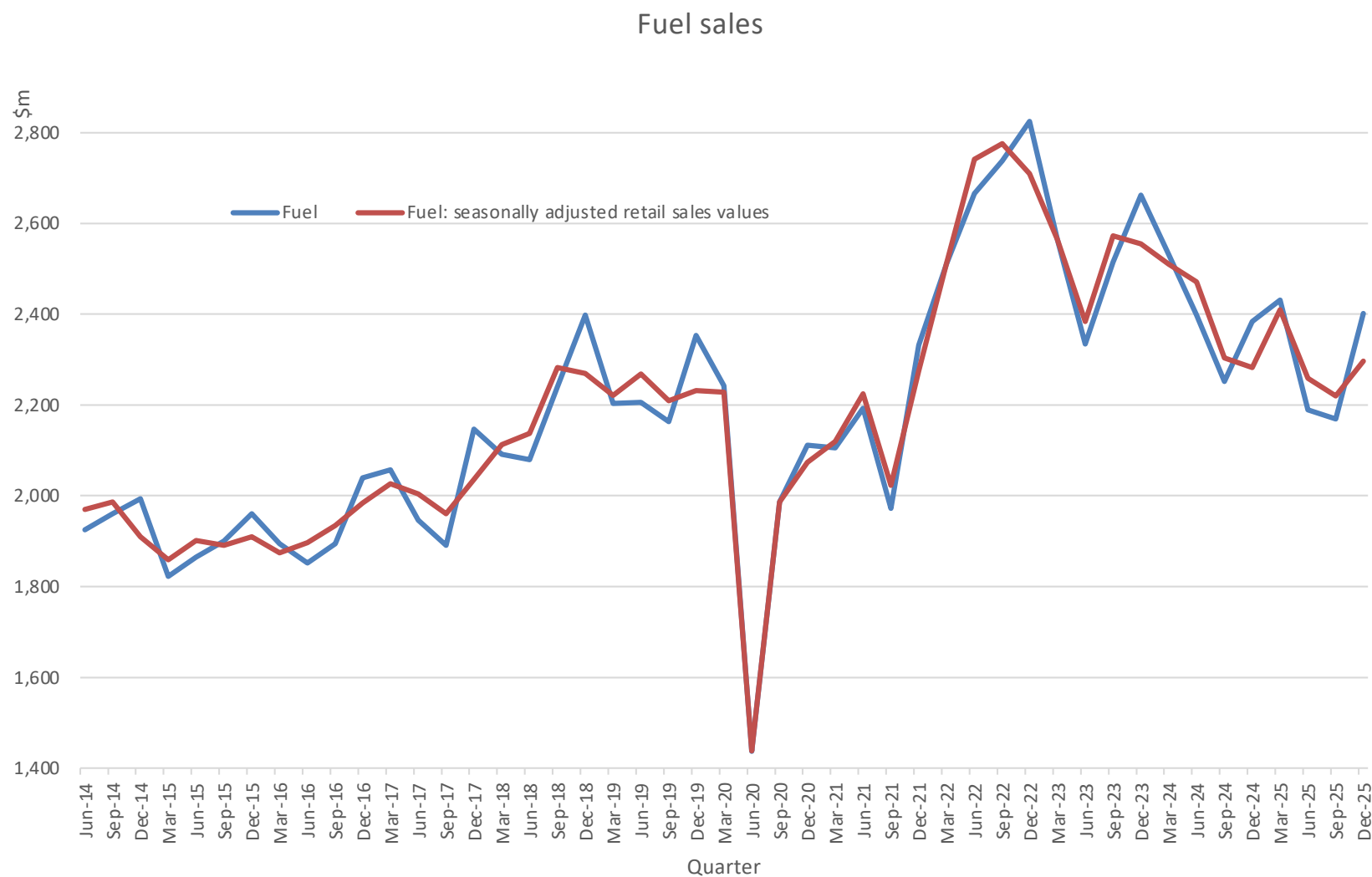
Seasonally adjusted retail sales volumes by industry, change from December 2019 quarter to March 2020 quarter



# NZ Retail Survey: Fuel

June 2014 to December 2025

Statistics NZ: various retail trade survey reports



<https://www.stats.govt.nz/information-releases/retail-trade-survey-december-2025-quarter/>

A screenshot of a Microsoft Excel warning dialog box. At the top is the Excel logo, a green square with a white 'X'. Below the logo, the text reads: "This workbook contains links to one or more external sources that could be unsafe." in bold black font. Underneath this, a paragraph explains: "If you trust the links, update them to get the latest data. Otherwise, you can keep working with the data you have." At the bottom, there are two buttons: a light gray button labeled "Don't Update" and a green button labeled "Update".

### Task 1: Line Chart: Supermarket sales values Dec23-Dec25

THE ENGINE  
OF THE **NEW**  
NEW ZEALAND

# NZ Retail Survey: Tables

December 2023 to December 2025

## Task 1: Line Chart: Supermarket sales values Dec23-Dec25

Suggestion: copy cells E12:U12 to the 'Workings tab' and follow instructions there

|    | A                                                                                                                                                                                                                            | B                                                                   | C                                                                                                                      | D | E      | F | G      | H | I      | J |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|---|--------|---|--------|---|--------|---|
| 1  |                                                                                                                                                                                                                              |                                                                     | PASTE ALL REQUESTED DATA HERE, then delete the blank (formatting) columns D, F, H... to insert individual line charts. |   |        |   |        |   |        |   |
| 2  | Location:                                                                                                                                                                                                                    |                                                                     | Dec-23                                                                                                                 |   | Mar-24 |   | Jun-24 |   | Sep-24 |   |
| 3  | Task 1: Table 1 row 12                                                                                                                                                                                                       | Actual Supermarket sales values Dec23-Dec25                         | 7,076                                                                                                                  |   |        |   |        |   |        |   |
| 4  | Task 2: Table 1 row 31                                                                                                                                                                                                       | Supermarket % change on same quarter previous year                  |                                                                                                                        |   |        |   |        |   |        |   |
| 5  | Task 5: Table 2 row 95                                                                                                                                                                                                       | Supermarket sales volumes (3Q 2010 prices) Dec23-Dec25              |                                                                                                                        |   |        |   |        |   |        |   |
| 6  | Task 6 Table 2 row 114                                                                                                                                                                                                       | Supermarket % change on same quarter previous year (3Q 2010 prices) |                                                                                                                        |   |        |   |        |   |        |   |
| 7  | Create graphs from each row, in separate charts                                                                                                                                                                              |                                                                     |                                                                                                                        |   |        |   |        |   |        |   |
| 8  | You can create one 'pretty' graph as per below, then copy and paste that chart and amend the row the chart is pulling data from (eg from C3:S3 for actual sales values, to C4:S4 for % change) - use the formula bar to edit |                                                                     |                                                                                                                        |   |        |   |        |   |        |   |
| 9  | Make graphs 'pretty' with axis titles - can remove legend if this duplicates chart title                                                                                                                                     |                                                                     |                                                                                                                        |   |        |   |        |   |        |   |
| 10 | Hint: make sure the X-axis shows the quarters ending March, June.... Excel will default to 2-monthly axis labels.                                                                                                            |                                                                     |                                                                                                                        |   |        |   |        |   |        |   |
| 11 | → To change axis labels: double-click the axis text, choose Axis Options and change Major units to 3 monthly                                                                                                                 |                                                                     |                                                                                                                        |   |        |   |        |   |        |   |
| 12 |                                                                                                                                                                                                                              |                                                                     |                                                                                                                        |   |        |   |        |   |        |   |
| 13 | → To add axis titles: click on the chart and choose Chart Tools > Design menu option; click on Add Chart Element > Primary Vertical.                                                                                         |                                                                     |                                                                                                                        |   |        |   |        |   |        |   |
| 14 | → Despite the text being side-ways, you can edit it normally by double-clicking and typing to replace.                                                                                                                       |                                                                     |                                                                                                                        |   |        |   |        |   |        |   |
| 15 | Example: raw chart and some modifications:                                                                                                                                                                                   |                                                                     |                                                                                                                        |   |        |   |        |   |        |   |

**Bounds**

Minimum 01/12/2023

Maximum 01/12/2025

**Units**

Major 3 Months

Minor 1 Months

Lab 3 Workings IRI SKUs Lab 4 SustainabilityData

**HERE**

**Andrew Murphy:**  
Hint: hold down the control / command key as you click ONCE on each column header (column letter). You can select multiple columns, and delete them all at once.  
NB: if you select a wrong column, you can now deselect it by clicking on the wrong cell / column WHILE STILL HOLDING the control/command key

## Task 2: Line Chart: Supermarket % change on same quarter previous year

Suggestion: copy cells E31:U31 to the 'Workings tab' and follow instructions there

# NZ Retail Survey: Percentage Change

## December 2023 to December 2025

Be careful with pasting: copy across first (simple paste)  
Then copy cells G51:U51 and use Paste Special > Formula  
for cells G52:U66

|    |                                                                       |             |      |      |  |  |  |  |  |  |
|----|-----------------------------------------------------------------------|-------------|------|------|--|--|--|--|--|--|
| 49 | <b>Task 3: Complete Percentage Change on previous quarter: Values</b> |             |      |      |  |  |  |  |  |  |
| 50 | Percentage change from previous quarter                               |             |      |      |  |  |  |  |  |  |
| 51 | Supermarket and grocery stores                                        | 10.1        | -4.0 | -4.2 |  |  |  |  |  |  |
| 52 | Specialised food                                                      | 16.2        |      |      |  |  |  |  |  |  |
| 53 | Liquor                                                                | 22.0        |      |      |  |  |  |  |  |  |
| 54 | Non-store and commission-based retailing                              | 17.3        |      |      |  |  |  |  |  |  |
| 55 | Department stores                                                     | 34.4        |      |      |  |  |  |  |  |  |
| 56 | Furniture, floor coverings, houseware, textiles                       | 10.9        |      |      |  |  |  |  |  |  |
| 57 | Hardware, building, and garden supplies                               | 8.1         |      |      |  |  |  |  |  |  |
| 58 | Recreational goods                                                    | 33.3        |      |      |  |  |  |  |  |  |
| 59 | Clothing, footwear, and accessories                                   | 28.8        |      |      |  |  |  |  |  |  |
| 60 | Electrical and electronic goods                                       | 12.4        |      |      |  |  |  |  |  |  |
| 61 | Pharmaceutical and other store-based retailing                        | 17.1        |      |      |  |  |  |  |  |  |
| 62 | Accommodation                                                         | 17.4        |      |      |  |  |  |  |  |  |
| 63 | Food and beverage services                                            | 6.9         |      |      |  |  |  |  |  |  |
| 64 | <b>Core industries total</b>                                          | <b>14.4</b> |      |      |  |  |  |  |  |  |
| 65 | Motor vehicles and parts                                              | -3.5        |      |      |  |  |  |  |  |  |
| 66 | Fuel                                                                  | 5.8         |      |      |  |  |  |  |  |  |
| 67 | <b>All industries total</b>                                           | <b>11.2</b> |      |      |  |  |  |  |  |  |

Which is a better indicator of central tendency for this data: mean or median? (overall)

|    |                                                               |      |  |  |  |  |  |  |  |  |
|----|---------------------------------------------------------------|------|--|--|--|--|--|--|--|--|
| 69 | <b>Task 4: Complete Percentage of "Core Industries" total</b> |      |  |  |  |  |  |  |  |  |
| 70 | Percentage of Core Industries Total                           |      |  |  |  |  |  |  |  |  |
| 71 | Supermarket and grocery stores                                | 27.2 |  |  |  |  |  |  |  |  |
| 72 | Specialised food                                              |      |  |  |  |  |  |  |  |  |
| 73 | Liquor                                                        |      |  |  |  |  |  |  |  |  |

# NZ Retail Survey: Constant Dollars

## December 2023 to December 2025

|     |                                                                                               |               |               |               |               |               |               |               |               |               |
|-----|-----------------------------------------------------------------------------------------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| 89  | <b>Actual retail sales volumes (expressed in September 2010 quarter prices)<sup>(1)</sup></b> |               |               |               |               |               |               |               |               |               |
| 90  | By quarter by industry                                                                        |               |               |               |               |               |               |               |               |               |
| 91  |                                                                                               |               |               |               |               |               |               |               |               |               |
| 92  |                                                                                               | 2022          | 2023          |               |               |               | 2024          |               |               |               |
| 93  | Industry                                                                                      | Dec           | Mar           | Jun           | Sep           | Dec           | Mar           | June          | Sep           | Dec           |
| 95  | Supermarket and grocery stores                                                                | 5,054         | 4,795         | 4,605         | 4,598         | 5,075         | 4,788         | 4,592         | 4,555         | 4,990         |
| 96  | Specialised food                                                                              | 387           | 340           | 339           | 321           | 392           | 351           | 327           | 319           | 397           |
| 97  | Liquor                                                                                        | 485           | 441           | 396           | 394           | 481           | 407           | 378           | 369           | 472           |
| 98  | Non-store and commission-based retailing                                                      | 553           | 445           | 434           | 453           | 499           | 427           | 447           | 447           | 515           |
| 99  | Department stores                                                                             | 2,360         | 1,704         | 1,855         | 1,679         | 2,368         | 1,738         | 1,912         | 1,806         | 2,423         |
| 100 | Furniture, floor coverings, houseware, textiles                                               | 724           | 621           | 632           | 639           | 741           | 625           | 644           | 668           | 764           |
| 101 | Hardware, building, and garden supplies                                                       | 2,293         | 1,977         | 1,983         | 2,018         | 2,240         | 1,932         | 1,944         | 2,022         | 2,291         |
| 102 | Recreational goods                                                                            | 884           | 729           | 653           | 606           | 911           | 740           | 695           | 655           | 994           |
| 103 | Clothing, footwear, and accessories                                                           | 1,394         | 1,106         | 1,134         | 1,023         | 1,378         | 1,128         | 1,114         | 1,004         | 1,363         |
| 104 | Electrical and electronic goods                                                               | 2,305         | 1,992         | 2,014         | 2,042         | 2,403         | 2,062         | 2,320         | 2,517         | 2,879         |
| 105 | Pharmaceutical and other store-based retailing                                                | 1,868         | 1,656         | 1,675         | 1,622         | 1,816         | 1,674         | 1,680         | 1,689         | 2,037         |
| 106 | Accommodation                                                                                 | 851           | 951           | 729           | 685           | 886           | 973           | 759           | 722           | 933           |
| 107 | Food and beverage services                                                                    | 2,713         | 2,580         | 2,468         | 2,460         | 2,718         | 2,550         | 2,473         | 2,545         | 2,767         |
| 108 | <b>Core industries total</b>                                                                  | <b>21,873</b> | <b>19,338</b> | <b>18,917</b> | <b>18,540</b> | <b>21,908</b> | <b>19,394</b> | <b>19,282</b> | <b>19,316</b> | <b>22,826</b> |
| 109 | Motor vehicles and parts                                                                      | 3,464         | 3,502         | 3,409         | 3,555         | 3,516         | 3,624         | 3,623         | 3,883         | 3,816         |
| 110 | Fuel                                                                                          | 1,537         | 1,496         | 1,416         | 1,417         | 1,525         | 1,482         | 1,396         | 1,378         | 1,481         |
| 111 | <b>All industries total</b>                                                                   | <b>26,875</b> | <b>24,336</b> | <b>23,743</b> | <b>23,512</b> | <b>26,949</b> | <b>24,500</b> | <b>24,301</b> | <b>24,577</b> | <b>28,122</b> |

**Task 5: Line Chart: Supermarket sales volumes (3Q 2010 prices) Dec23-Dec25**

(see workings tab)

**Task 6: Line Chart: Supermarket % change on same quarter previous year (3Q Sep10 prices)**

**132 Task 7: Complete Percentage Change on previous quarter: Volumes**

|              |          |          |       |                    |
|--------------|----------|----------|-------|--------------------|
| <b>Lab 3</b> | Workings | IRI SKUs | Lab 4 | SustainabilityData |
|--------------|----------|----------|-------|--------------------|

**153 Task 8: In the IRI SKUs sheet:**

**154** Data obtained from the Temple plugin for Excel, provided by IRI. See instructions in Stream Topic 3 for how to connect to the Temple plugin for Excel.

**155** See instructions in Stream Topic 3 for how extract information using the Temple plugin for Excel. (We will use this tool in future labs, so good to practice)

# Lab 3: IRI widgets dataset: SKUs

- 153 **Task 8: In the IRI SKUs sheet:**
- 154 Data obtained from the Temple plugin for Excel, provided by IRI. See instructions in Stream Topic 3 for how to connect to the Temple plugin for Excel if you wish to replicate t
- 155 See instructions in Stream Topic 3 for how extract information using the Temple plugin for Excel . (We will use this tool in future labs, so good to practice now.)
- 156 **1) Sort the data in the IRI SKUs sheet by sales Dollars (column B)** **Murphy, Andrew:** Products should be at the top, after Total Widgets).
- 157 Many of the products show no sales (\$0 and 0 sales units) so these **Hint: select the whole row:** B: you could delete them instead, but this screws up the cell ref
- 158 **2) In column M, create cumulative sales dollars by progressively** click on row 7 and select the previous rows (ignoring Total Widgets)
- 159 In cell M7, insert the formula =B7. In cell M8, insert the formula = running (cumulative) total of sales dollars.
- 160 **3) How many products have \$0 sales recorded for the period?** down to row 85
- 161 Create a line chart (cumulative sales: column M; leave the X-axis l
- 162 **3) In IRI SKUs cells C88 & C89, calculate mean & median units sold** Mean: 0 Median: 0
- 164 **4) In cells E88 & E89, calculate mean & median price per unit, and copy here:** Mean: 0.00 Median: 0.00
- 165 **5) Why does the mean price you just calculated differ from the value in cell E6 (Total Widgets price per unit)?**
- 166
- 167
- 168 **6) How many products are in the above mean price set?** 0 **How many are in the below mean price set?** 0
- 170 **7) How many products are in the above median price set?** 0 **How many are in the below median price set?** 0
- 171 **8) Calculate total units sold for products that are above vs below mean price** Total units sold above: 0 Total units sold below: -
- 172 **9) Calculate total sales dollars for products that are above vs below mean price** Total dollars above: 0 Total dollars below: -
- 173
- 174 Chart of cumulative sales dollars:
- 175

(all those red triangles are hints!)

|    | A                                         | B | C       | D     | E                                                                   | F             | G | H | I | J                                                           | K | L | M | N |
|----|-------------------------------------------|---|---------|-------|---------------------------------------------------------------------|---------------|---|---|---|-------------------------------------------------------------|---|---|---|---|
| 1  | <b>IRI SKUs Ranking Report - All SKUs</b> |   |         |       |                                                                     |               |   |   |   |                                                             |   |   |   |   |
| 2  | Data retrieved via Temple plugin          |   |         |       |                                                                     |               |   |   |   |                                                             |   |   |   |   |
| 3  |                                           |   |         |       |                                                                     |               |   |   |   |                                                             |   |   |   |   |
| 4  | MAT To 19/06/22 New Zealand Grocery       |   |         |       |                                                                     |               |   |   |   |                                                             |   |   |   |   |
| 87 |                                           |   |         | Units |                                                                     | Average Price |   |   |   |                                                             |   |   |   |   |
| 88 |                                           |   | Mean:   |       |                                                                     | Mean:         |   |   |   |                                                             |   |   |   |   |
| 89 |                                           |   | Median: |       |                                                                     | Median:       |   |   |   |                                                             |   |   |   |   |
| 90 |                                           |   |         |       |                                                                     |               |   |   |   |                                                             |   |   |   |   |
| 91 |                                           |   |         |       |                                                                     |               |   |   |   |                                                             |   |   |   |   |
| 92 | Total units sold above mean price:        |   |         |       | Number of products above mean price:                                |               |   |   |   | Number of products above median price:                      |   |   |   |   |
| 93 | Total units sold below mean price:        |   |         |       | Number of product below mean price:                                 |               |   |   |   | Number of products below median price:                      |   |   |   |   |
| 94 |                                           |   |         |       |                                                                     |               |   |   |   |                                                             |   |   |   |   |
| 95 | Total dollar sales above mean price:      |   |         |       |                                                                     |               |   |   |   | *The alternative, but more clunky way, is to sort the E col |   |   |   |   |
| 96 | Total dollar sales below mean price:      |   |         |       | Note these cells are locted in a different order on the Lab 3 sheet |               |   |   |   |                                                             |   |   |   |   |

# Lab 3: AI Declaration

200 <<End of Lab 3>>

201

## 202 AI Declaration for Lab 3:

203 ☐ I have not used any AI tools or technologies to prepare this assessment

204 OR

205 ☐ I acknowledge the use of [AI tool or technology name and link] to generate [summary of what was generated]:

206

*AI tool used:*

207

*I entered the following prompt(s):*

208

209

210

*I used the output to / I modified the output to: [expand as necessary]*

211

212

213

214

215

216